Robust Local-Unitary Rigidity of Stabilizer AME States

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Abstract

Let $q = p^e$ and $m \geq 2$. We prove that every product unitary mapping one stabilizer $\mathrm{AME}(2m, q)$ state to another is Clifford on each party, also when party relabelling is allowed. The result covers arbitrary additive stabilizers: local Clifford actions lie in $\mathrm{Sp}_{2e}(\mathbb{F}_p)$, with no $\mathbb{F}_q$-linearity assumption. The proof is a support count. Stabilizers contained in any $(m+1)$-party set project bijectively onto every retained party's Weyl labels, so the reduced operator determines all local Weyl axes.

We also prove robust rigidity. Below an explicit state-vector defect threshold of order $\min\{q^{-1/2}, (2m)^{-1/2}\}$, a product unitary decomposes into an exact symmetry and a residual whose collective generator norm is at most $\pi\sqrt{q}\,\varepsilon$. Each local factor first rounds to a Clifford within normalized Hilbert–Schmidt distance 8ε. Three-region cleaning and Weyl–Fourier concentration give these frames; quantized stabilizer overlaps and a balanced-cut estimate control the residual. The rounded frames satisfy all transition equations already for $\varepsilon < \sqrt{2}/68$, independently of party count and local dimension, while a stabilizer-character correction remains uncontrolled at that larger radius.

The exact proof also gives a complete finite transition-map invariant and factorwise Clifford rigidity for transversal encoder conversions. For promised stabilizer-AME check matrices, fixed-label recognition returns compact symplectic witnesses in polynomial time in m for fixed q; stabilizer phases additionally determine a Clifford–Pauli conversion. Selected marginal tests give fidelity certificates and an observable sufficient condition for the robust theorem.

Index Terms—Absolutely maximally entangled states, local Clifford group, local-unitary equivalence, quantitative rigidity, quantum MDS codes, stabilizer cleaning, transversal gates, Weyl operators.

1 Introduction

An absolutely maximally entangled state has maximally mixed reductions on every set containing at most half of its parties. Such states are also called perfect tensors and, after one party is designated as an input, give quantum maximum-distance-separable encoders. They therefore sit at the intersection of multipartite entanglement, stabilizer codes, transversal gates, and tensor-network constructions. Operationally, AME states serve as resources for quantum error correction, threshold secret sharing, and multi-user teleportation; a recent survey reviews these applications and the known local-unitary classes [RMBR+26].

This paper asks how rigid their local coordinates are. With the party labels fixed, two states are *local-unitary (LU) equivalent* if a tensor product of one-party unitaries maps one state ray to the other. We say *LU up to party relabelling* when a party permutation is also allowed. The analogous relations are *local-Clifford (LC) equivalence* and LC equivalence up to party relabelling, with every one-party factor required to normalize the chosen Weyl, or Pauli, system. The theorem below uses the latter convention; setting the permutation to the identity gives the fixed-label statement. Local-Clifford equivalence is discrete and can be studied through finite symplectic linear algebra; local-unitary equivalence is a priori continuous and much larger. The two relations do not coincide for general stabilizer states [JCWY10].

The distinction is both structural and operational. LU equivalence asks whether two preparations represent the same multipartite entanglement resource after independent changes of local basis. LC equivalence reduces that question to finite symplectic data and is directly compatible with the stabilizer description used for encoding and fault-tolerant gates. For an AME state viewed as an encoder Choi state, the same question asks whether a transversal conversion can contain a genuinely non-Clifford factor. A quantitative version is needed when the conversion or symmetry is only approximately realized.

Within the stabilizer AME class on an even number of at least four parties, maximal entanglement removes that difference.

For $q = p^e$, "Clifford" below always means the normalizer of the additive Weyl system. Its action on one-party labels lies in $\mathrm{Sp}_{2e}(\mathbb{F}_p)$; it need not be an $\mathbb{F}_q$-linear 2×2 transformation.

Theorem 1.1 (Local-unitary rigidity)**.** Let $q = p^e$, let $m \geq 2$, and let $|\psi\rangle, |\phi\rangle$ be stabilizer AME$(2m, q)$ states for the standard additive Weyl system. Suppose that, for a party permutation π, single-qudit unitaries $U_1, \dots, U_{2m}$, and a phase θ,

$$(U_1 \otimes \cdots \otimes U_{2m})\pi|\psi\rangle = e^{i\theta}|\phi\rangle.$$

Then every U_i normalizes the Weyl system. Thus the displayed equivalence is local Clifford.

The theorem converts a continuous equivalence problem into a finite one. It is valid for every prime power, for arbitrary additive stabilizers, and does not assume CSS form, equal stabilizer phases, classical linearity over $\mathbb{F}_q$, or a particular construction of the AME state. The restriction $m \geq 2$ is sharp: a Bell pair has the continuous symmetry $U \otimes \overline{U}$.

The proof uses one elementary support calculation. On any $(m+1)$-party set, the stabilizer labels supported there form a group of order q^2. Projection onto the Weyl-label space $\mathbb{F}_q^2$ of any retained party is bijective; this space has dimension $2e$ over $\mathbb{F}_p$. Consequently the reduced operator contains every local Weyl axis exactly once in a matched diagonal decomposition. The uniqueness of that decomposition forces each local unitary to permute the Weyl axes and hence to be Clifford. In ordinary language,

$$\text{AME support counting} \implies \text{all local Weyl axes appear} \implies \text{every local factor is Clifford.} \tag{1.1}$$

Quantitative question

Exact rigidity does not by itself say what happens when a product unitary only approximately preserves the state. For $U = \bigotimes_i U_i$, define its state-vector defect by

$$\varepsilon(U) = \min_{|z|=1} \| U|\psi\rangle - z|\psi\rangle \|.$$

The direct marginal proof is poorly conditioned for this problem: the nonidentity part of one minimum-support marginal has exponentially small Hilbert–Schmidt norm. We instead regard each party in turn as the logical leg of the associated $[[2m-1, 1, m]]_q$ code. Its physical parties can be partitioned into three correctable regions. A leakage-aware cleaning argument tests the induced logical action by nested commutators, and a finite Fourier argument rounds that action to a Clifford.

The exact constants are useful for applications, so we state them once in the introduction. Their scaling and proof mechanism are the main points. Write $G(\psi)$ for the group of product unitaries that preserve the ray of $|\psi\rangle$.

Theorem 1.2 (Quantitative rounding)**.** Let $|\psi\rangle$ be a stabilizer AME(n,q) state with $n = 2m$, $m \geq 2$, and let $U = \bigotimes_i U_i$ be a product unitary. Put

$$R_{n,q}^{\text{clean}} = \min\left\{\frac{1}{4\sqrt{2q}}, \frac{1}{8\pi\sqrt{n}}\right\}. \tag{1.2}$$

If $\varepsilon(U) < R_{n,q}^{\text{clean}}$, then, up to phase,

$$U = g\bigotimes_{i=1}^{n} e^{ih_i}, \qquad g \in G(\psi), \qquad \left(\sum_i \|h_i\|_F^2\right)^{1/2} \leq \pi\sqrt{q}\,\varepsilon(U), \tag{1.3}$$

where the h_i are traceless Hermitian with spectral spread $\lambda_{\max}(h_i) - \lambda_{\min}(h_i)$ at most π. Before the exact symmetry g is selected, each U_i is within normalized Hilbert–Schmidt distance $8\varepsilon(U)$ of a one-qudit Clifford. Uniformly over prime powers and existing AME states,

$$R_{n,q}^{\text{clean}} = \Theta\Big(\min\{q^{-1/2}, n^{-1/2}\}\Big).$$

The proof has three stages. First, quantitative cleaning places every nested logical Weyl commutator within 8ε of a scalar. Second, Fourier concentration on the finite Weyl group produces a nearby local Clifford at each party. Third, the quantized overlap of stabilizer states separates distinct product-Clifford branches by the uniform gap $d_p = (2 - 2p^{-1/2})^{1/2}$. The AME second moment bounds the distance to the rounded Clifford branch; a balanced-cut estimate using minimum-support labels then gives the collective residual bound in (1.3). The radius is a certified lower bound, not an optimality claim.

Here approximation refers to the state-vector defect of a product unitary acting on an exact code state. This differs from approximate Eastin–Knill results, where error correction itself is relaxed; see, for example, [Ale26]. The two questions share the same fault-tolerance motivation but control different errors.

An exact product-unitary intertwiner between two states transfers this result without changing the constants (Corollary 6.8).

The two headline proof branches are independent. Support geometry first recovers every Weyl axis, which yields exact rigidity and then the transition-map and endomorphism structure. Separately, the same support geometry gives correctable regions, which feed the robust rounding theorem. Appendix B isolates the weaker consequences of pairwise maximal mixing alone and is not used to prove either branch.

Consequences of the exact theorem

The operational code consequence follows by treating one party as the logical input of the associated quantum MDS encoder. Throughout, *transversal* means a tensor product of one-site physical operators with the physical coordinates fixed. Such gates limit the direct spread of errors between physical subsystems, which makes them central to fault-tolerant constructions. The Eastin–Knill theorem rules out a universal transversal gate set for an exact finite-dimensional code [EK09]; the natural family-specific question is therefore which transversal gates remain.

Corollary 1.3 (Factorwise transversal rigidity)**.** Let $m \geq 2$, and let $V_\psi, V_\phi : \mathbb{C}^q \to (\mathbb{C}^q)^{\otimes(2m-1)}$ be encoders whose normalized Choi states are stabilizer AME$(2m,q)$ states. If $U_i \in U(q)$ and

$$(U_1 \otimes \cdots \otimes U_{2m-1})V_\psi = V_\phi L,$$

then L and every U_i are Clifford. In particular, no code in this family admits a transversal non-Clifford logical unitary.

General cleaning arguments already constrain the *logical* action of a transversal gate [PY15, Lemma 5]. Corollary 1.3 is stronger in a different direction: it identifies every physical one-site factor as Clifford as well as the induced logical gate. This factorwise conclusion is what turns LU equivalence of the Choi states into finite local symplectic data.

Transition maps. The support bijections retain how the local Weyl frames are related. For an $(m+1)$-party support A, projection at party i gives a bijection $p_i : L_\psi(A) \to \mathbb{F}_q^2$. The maps

$$M_A(i,j) = p_j p_i^{-1}$$

are the usual operator-pushing maps between the local Weyl-label spaces. Their compositions around closed paths are holonomies.

Theorem 1.4 (Classification by minimum-support transition maps)**.** Let $m \geq 2$. After a fixed party relabelling, two stabilizer $\mathrm{AME}(2m,q)$ states are LU equivalent exactly when their systems of minimum-support transition maps are equivalent by local trace-symplectic changes of Weyl frame. For one state, if $\mathrm{Aut}_{\mathrm{tr}}(\psi)$ is the group of compatible local symplectic automorphisms of the transition system and Γ_ψ its fixed-party projective product-unitary symmetry group, then

$$1 \longrightarrow L_\psi \longrightarrow \Gamma_\psi \longrightarrow \mathrm{Aut}_{\mathrm{tr}}(\psi) \longrightarrow 1$$

is exact. When q is prime, $\mathrm{Aut}_{\mathrm{tr}}(\psi)$ is the determinant-one unit group of the local endomorphism algebra described in Section 5.

The invariant is also effective. After fixing any half-set of parties, only the m minimum supports obtained by adjoining one complementary party are needed: their supported label groups form a direct sum equal to the full stabilizer label space, and no smaller collection of minimum-support subgroups can span it (Proposition 4.1). Fixed-label recognition then enumerates one base symplectic frame and checks each candidate by finite linear algebra. In particular, it is polynomial in m for fixed local dimension. Corollary 4.3 gives the explicit candidate count and the separate cost of unknown party relabelling. The input is promised stabilizer-AME check matrices with fixed party labels; the output is a compact symplectic witness, upgraded to a Clifford–Pauli conversion when stabilizer phases are supplied. These data also determine an intrinsic endomorphism algebra in prime dimension (Section 5), certify the state from m minimum-support marginals, and decide stochastic local conversion (Section 4.2). Complementary marginal tests also give a fidelity certificate with gap $(m+1)/(2m)$, supplying an observable sufficient condition for robust rigidity when the preparation is promised to be a product-unitary image (Section 4.3). These consequences are optional on a first reading of the two rigidity proofs.

At the radius

$$R^{\mathrm{trans}} = \frac{\sqrt{2}}{68}, \tag{1.4}$$

independent of both party count and local dimension, the symplectic maps recovered by cleaning already preserve every transition map exactly. This does not yet identify the exact product symmetry. Localized commutators cancel Weyl-translation phases, so they determine the stabilizer label space but not its character. The resulting product-Pauli correction need not be close to the rounded product Clifford. We refer to this as the unresolved stabilizer-character, or product-Pauli phase, correction. Controlling it is the missing ingredient in a party-count-independent global entry radius.

Relation to prior work and contribution

The exact three-cleanable-region Clifford obstruction is due to Pastawski and Yoshida [PY15, Lemma 5], building on stabilizer cleaning [BT09, Lemma 1] and the localization method behind transversal-gate hierarchy bounds [BK13, JOKY18]. We use only the exact cleaning and localization consequences and prove the leakage estimate, finite Weyl–Fourier rounding, and explicit AME composition here.

For exact LU rigidity, Rains recovered the three Pauli axes from a diagonal correlation tensor, and Van den Nest, Dehaene, and De Moor used that mechanism in a minimum-support criterion for qubit stabilizer states [Rai99, Theorem 13] [VdNDDM05, Theorem 1 and Corollary 1]. Under their qubit minimum-support hypotheses, their proof already makes the factors of the given LU Clifford. Our extension is the automatic complete-Weyl-support condition for arbitrary additive prime-power AME states, together with the constructive and quantitative conclusions. Subsequent work developed further sufficient conditions and exposed the limits of the LU–LC conjecture [ZCCC07, GVdN08, JCWY10]. Englbrecht and Kraus classify all qubit stabilizer local symmetries [EK20]. More recently, Claudet and Perdrix gave a graphical characterization using generalized local complementation and a quasi-polynomial decision algorithm, while Burchardt, de Jong, and Vandré reduced verification to modular linear equations [CP25b, CP25a, BdJV24]. The general qubit problem is now sharply separated from the AME case: LU and LC agree through 26 qubits but already differ at 27 [Cla26]. The present support argument instead uses the AME condition to extend the axis-recovery mechanism to arbitrary additive prime-power stabilizers. Because every minimum support contains the full local Weyl basis, no additional minimum-support hypothesis is required.

Tan proves the general AME–quantum-MDS orbit and symmetry–transversal correspondences [Tan26b, Theorems 3.5 and 3.6] and computes the four-qutrit symmetry and transversal groups [Tan26b, Theorem 5.3]; separate work classifies five-qubit AME states [Tan26a]. To our knowledge, the missing uniform statement was that every factor in an arbitrary additive prime-power stabilizer-AME conversion is Clifford, at every even length for which such a state exists. Our exact result supplies that factorwise statement, a complete transition-map invariant, and the finite recognition bound above. The same half-set geometry shows that m reductions to $m+1$ parties determine the state among all density operators, with both numbers optimal and an explicit parent Hamiltonian, and that the unitarity hypothesis can be dropped: every invertible product operator between two such states is Clifford up to scalars (Section 4.2). The first statement sharpens the generator-support criterion for qubit stabilizer states [WYW$^+$15, Theorem 2]; the second strengthens the Kempf–Ness equality of SLOCC and LU classes for AME states [BR20, Corollary 1]. The exact result also identifies, in prime dimension, the intrinsic endomorphism algebra behind the symmetry calculation. To our knowledge, no earlier result forces this algebra to be nonscalar for every four- or six-party stabilizer AME state; Tan's four-qutrit computation is an important special case. The quantitative result supplies an explicit neighborhood in which an approximate product symmetry can be assigned to an exact Clifford branch, while distinguishing symplectic compatibility from the remaining stabilizer-character correction. No new qualitative transversal-hierarchy obstruction is claimed.

The class continues to acquire nontrivial examples: recent self-dual MDS constructions establish stabilizer $\mathrm{AME}(12,5)$, $\mathrm{AME}(18,11)$, and $\mathrm{AME}(18,13)$ states [BB26]. Our results apply to such states without selecting a CSS, graph-state, or $\mathbb{F}_q$-linear presentation. Recent work classifies diagonal transversal Clifford groups and characterizes code-preserving Clifford circuits for broader binary code families and circuit models [DB25, Alb26]; here the input is instead the maximal entanglement of one stabilizer state, and the conclusion begins with an arbitrary product unitary.

The stabilizer assumption is essential to the discrete transition system. General AME families can contain infinitely many LU classes; Ramadas and Lakshminarayan give an orthogonal-array criterion producing such families [RL25, Theorem 1 and Corollary 1]. Perfect-tensor operator pushing and the encoder interpretation are used explicitly in holographic codes [PYHP15, Eq. (2.3) and Section 5.7].

Reading the paper

For the two main results, read the support dictionary in Section 2, the exact proof in Section 3, and the robust proof in Section 6. The four-qutrit example in Section 4.1 makes the support, frame and character data explicit. Its marginal tests in Section 4.3 lead to the observable rounding criterion in Corollary 6.11.

The finite classification and encoder consequences occupy the rest of Section 4; Section 5 refines the classification in prime dimension. These sections can be skipped on the first route. The appendices give the associated code structures, stabilizer-independent infinitesimal estimates, weighted and nonstabilizer marginal-test refinements, and the formal trust boundary. They are not needed for either headline rigidity theorem.

2 Stabilizer AME support geometry

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x + a\rangle, \qquad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes $\mathbb{C}W_v$.

The label-space commutator is measured by the *trace-symplectic pairing*

$$[v, w] = \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(ab' - ba') \in \mathbb{F}_p, \qquad v = (a, b), \quad w = (a', b').$$

After choosing dual $\mathbb{F}_p$-bases of $\mathbb{F}_q$, this Weyl system is the Pauli system of e p-level factors. Its projective Clifford normalizer has the standard exact sequence

$$1 \longrightarrow \mathbb{F}_p^{2e} \longrightarrow \mathrm{PCliff}(q) \longrightarrow \mathrm{Sp}_{2e}(\mathbb{F}_p) \longrightarrow 1, \tag{2.1}$$

including $p = 2$: the kernel is the projective Pauli group, and every trace-symplectic $\mathbb{F}_p$-linear label map has a Clifford lift [DDM03, HDDM05].

For n parties, a pure stabilizer state is fixed by an abelian Pauli group $\mathcal{S}$ of order q^n. Its projective label group $L \leq (\mathbb{F}_q^2)^n$, viewed additively over $\mathbb{F}_p$, is maximal isotropic for the trace-symplectic form. This is the standard arbitrary-additive prime-power stabilizer correspondence [KKKS06, Theorems 13 and 15]. For each $\ell \in L$, write the stabilizer lift as $\widetilde{W}_\ell = \alpha_\ell W_\ell$, where $\alpha_\ell \neq 0$. These phases encode the joint eigenvalue assignment on L, relative to the fixed Weyl convention; we call that assignment the *stabilizer character*. Then

$$|\psi\rangle\langle\psi| = q^{-n} \sum_{\ell \in L} \widetilde{W}_\ell. \tag{2.2}$$

If A is a party set, put

$$L(A) = \{\ell \in L : \operatorname{supp}(\ell) \subseteq A\}.$$

Weyl orthogonality and partial trace give

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L(A)} \widetilde{W}_\ell. \tag{2.3}$$

In particular, ρ_A^ψ is maximally mixed exactly when $L(A) = \{0\}$. Thus a stabilizer state on $2m$ parties is AME exactly when it has no nonidentity stabilizer label supported on at most m parties.

Equations (2.2)–(2.3) are the only state-specific input to the exact rigidity theorem. The same support geometry also identifies any chosen party as the logical leg of a one-qudit stabilizer encoder on the remaining $2m-1$ parties. The resulting code has parameters $[[2m-1,1,m]]_q$: erasure of at most $m-1$ physical parties is correctable because the complementary reductions of the AME state are maximally mixed. We use this encoder interpretation twice, first for the exact transversal consequence and then for quantitative cleaning.

3 Complete Weyl-basis marginals and LU rigidity

We first isolate the linear-algebra statement that recovers a diagonal tensor's coordinate axes. It is the complete-Weyl-basis counterpart of the three-Pauli argument in [Rai99, VdNDDM05]. The statement itself is classical: it is Jennrich's uniqueness theorem for trilinear decompositions, published in [Har70, §V] and there credited to Jennrich, whose conclusion that the factor matrices agree up to one permutation and compensating diagonal scalings is the monomial property below. It is the full-column-rank case of Kruskal's condition [Kru77], extended to arbitrarily many factors in [SB00]; see [KB09, §3.2] for the attribution and for a survey of the uniqueness theory. We give a short self-contained proof because the contraction argument is the one used in Proposition 3.2, and claim no novelty for either the statement or the proof.

Lemma 3.1 (Diagonal-tensor axes)**.** Let $r \geq 3$, let $E_1, \ldots, E_r$ be N-dimensional complex inner product spaces, and choose orthonormal bases $\{e_{ij} : 1 \leq j \leq N\}$. If

$$T = \sum_{j=1}^{N} \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \qquad \lambda_j \neq 0, \tag{3.1}$$

then the coordinate axes in every E_i are intrinsic to T. Consequently, a product of invertible linear maps carrying one tensor of the form (3.1) to another is monomial in the displayed bases.

Proof. For nonzero $x = \sum_j x_j e_{1j}^*$, contract the first factor against x. The result is

$$T_x = \sum_j \lambda_j x_j e_{2j} \otimes \cdots \otimes e_{rj}.$$

Flattening from E_2 to $E_3 \otimes \cdots \otimes E_r$ gives a matrix of rank $\#\{j : x_j \neq 0\}$. Thus T_x has rank one exactly when the ray of x is a dual coordinate axis. Product equivalence preserves this projective rank-one contraction locus. The inner products identify the recovered dual axes with the coordinate axes in the first factor. Apply the same argument in every factor. □

To avoid repeatedly stating the matched-basis condition, call an operator on $r \geq 3$ qudits *Weyl-diagonal with complete local support* if there is a q^2-element set P, a distinguished element $0 \in P$, and bijections $f_i : P \to \mathbb{F}_q^2$ with $f_i(0) = 0$, for which

$$R = \sum_{v \in P} \lambda_v \bigotimes_{i=1}^{r} W_{f_i(v)}, \qquad \lambda_v \neq 0. \tag{3.2}$$

The coefficients need not be equal. What matters is that the same q^2-element index set runs bijectively through the complete local Weyl basis at every party. Neither P nor the bijections need be $\mathbb{F}_q$-linear.

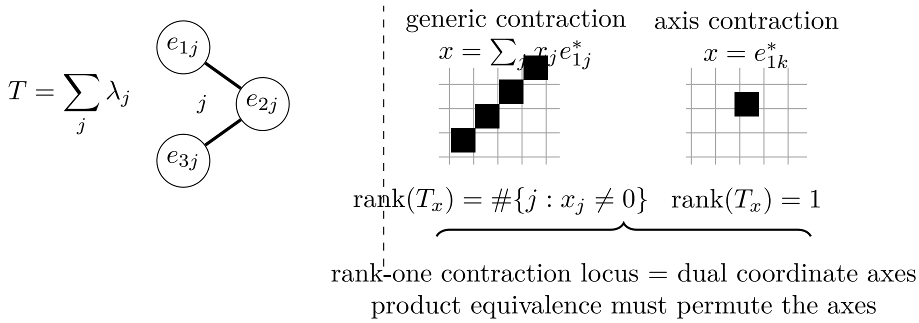

Figure 1: A diagonal tensor with all coefficients nonzero exposes its own axes. Contracting one leg against x and flattening the remainder gives rank equal to the number of nonzero coordinates of x, so the rank-one contraction locus is exactly the set of dual coordinate axes. For an $(m+1)$-party AME marginal in local Hilbert–Schmidt spaces, the recovered axes are the local Weyl axes, and a product conjugation must permute them.

Proposition 3.2 (Complete Weyl-basis marginal criterion)**.** Let R and R' be Weyl-diagonal operators with complete local support on the same $r \geq 3$ parties, possibly with different indexing sets, bijections, and coefficients. If $U_i \in U(q)$ and

$$R' = (U_1 \otimes \cdots \otimes U_r) R (U_1 \otimes \cdots \otimes U_r)^\dagger,$$

then every U_i is Clifford.

Proof. View R and R' as tensors in the local Hilbert–Schmidt spaces. After replacing W_v by the orthonormal basis $q^{-1/2}W_v$ and absorbing the common normalization into the coefficients, both have the form in Lemma 3.1, with $N = q^2$. Local conjugation acts invertibly on each factor, so it must permute the Weyl axes. It fixes the identity operator and hence the identity axis. It therefore permutes the nonidentity projective Weyl operators, which is exactly the single-qudit Clifford condition. □

Corollary 3.3 (Marginal-cover rigidity)**.** Let $|\psi\rangle, |\phi\rangle$ be n-qudit states and suppose a product unitary carries $|\psi\rangle$ to $|\phi\rangle$ up to phase, any party permutation having first been absorbed by relabelling $|\phi\rangle$, so that the two states are indexed by the same parties. If every party belongs to a set S of at least three parties for which both reduced operators ρ_S^ψ and ρ_S^ϕ are Weyl-diagonal with complete local support, then every local factor is Clifford.

Proof. Partial trace gives a product-unitary equivalence of the two reduced operators on each chosen S. Apply Proposition 3.2; the chosen sets cover all parties. □

Proposition 3.4 (Stabilizer-AME support theorem)**.** Let $|\psi\rangle$ be a stabilizer AME$(2m, q)$ state, and let A be any $m+1$ parties. Then

$$|L(A)| = q^2.$$

Every nonidentity label in $L(A)$ has support exactly A, and for each $i \in A$ the coordinate projection

$$p_i : L(A) \longrightarrow \mathbb{F}_q^2$$

is bijective.

Proof. The projective label group L has order q^{2m}. Restrict its labels to the $m-1$ parties outside A. The target has order $q^{2(m-1)}$, and the kernel is $L(A)$; hence

$$|L(A)| \geq q^{2m}/q^{2(m-1)} = q^2. \tag{3.3}$$

Fix $i \in A$. A label in the kernel of p_i is supported on the m-set $A \setminus \{i\}$. The AME condition and (2.3) exclude every such nonidentity stabilizer, so p_i is injective. Its codomain has order q^2, giving the reverse inequality in (3.3) and hence bijectivity. The same no-small-support argument shows that every nonidentity element of $L(A)$ has full support A. $\square$

Proposition 3.5 (A stabilizer-AME marginal fixes the local Weyl axes)**.** Let $|\psi\rangle, |\phi\rangle$ be stabilizer AME$(2m, q)$ states with $m \geq 2$. If a product unitary on an $(m+1)$-set A carries ρ_A^ψ to ρ_A^ϕ, then every local factor on A is Clifford.

Proof. By (2.3),

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L_\psi(A)} \widetilde{W}_\ell.$$

Index this sum by $P = L_\psi(A)$. Proposition 3.4 says that every coordinate projection $P \to \mathbb{F}_q^2$ is bijective. Relative to the fixed Weyl convention, each stabilizer lift contributes a nonzero coefficient. Thus ρ_A^ψ is Weyl-diagonal with complete local support in the sense of (3.2), and the same holds for ρ_A^ϕ. Apply Proposition 3.2. $\square$

Proof of Theorem 1.1. Absorb the party permutation by relabelling $|\phi\rangle$. Partial trace gives a product-unitary equivalence $\rho_A^\psi \sim \rho_A^\phi$ for every $(m+1)$-set A. Given a party i, choose such a set containing it and apply Proposition 3.5. Thus every U_i is Clifford. $\square$

4 Minimum-support transition maps and encoder consequences

The same support calculation recovers standard quantum-MDS structure. If $|A| \leq m$, the AME condition gives $L(A) = \{0\}$. If $|A| > m$, then $\rho_{A^c}^\psi$ is maximally mixed, so purity of the global state gives $\mathrm{Tr}((\rho_A^\psi)^2) = q^{-(2m-|A|)}$. On the other hand, (2.3) and Weyl orthogonality give $\mathrm{Tr}((\rho_A^\psi)^2) = q^{-|A|}|L(A)|$. Hence, for every party set A,

$$|L(A)| = q^{2\max(|A|-m,0)}. \tag{4.1}$$

Möbius inversion is the stabilizer-state specialization of the quantum-MDS Shor–Laflamme distribution of Huber and Grassl [HG20, Theorems 8 and 9]; in particular the number of weight-$m+1$ projective stabilizers is $\binom{2m}{m+1}(q^2-1)$. The following systematic form strengthens minimum-weight generation: recognition needs only m of these supports, and that number is optimal among spanning families of minimum-support subgroups. Section 4.1 works out these constructions for one four-qutrit state and can be read alongside the following definitions.

Proposition 4.1 (Half-set systematic decomposition)**.** Fix an m-party set B, and let $P_j = \mathbb{F}_q^2$ be the local label space at party j. The coordinate projection

$$\mathrm{pr}_{B^c} : L \longrightarrow \bigoplus_{j \notin B} P_j$$

is an isomorphism of additive groups. Hence there is a unique $\mathbb{F}_p$-linear map

$$K_B : \bigoplus_{j \notin B} P_j \longrightarrow \bigoplus_{i \in B} P_i$$

whose graph is L. For the direct-sum trace-symplectic forms it satisfies

$$[K_B x, K_B y]_B = -[x, y]_{B^c}.$$

After local $\mathbb{F}_p$-bases are chosen, row reduction therefore puts any row-generator check matrix for L, ordered as B, B^c, in systematic form $[K_B^{\mathsf{T}} \mid I]$; here K_B acts on column vectors. Moreover, if $I \subseteq B$, $J \subseteq B^c$, and $|I| = |J|$, then the party-block submatrix

$$K_{I,J} : \bigoplus_{j \in J} P_j \longrightarrow \bigoplus_{i \in I} P_i$$

is an isomorphism. Conversely, a maximal isotropic subspace of the form graph(K_B) is the label space of a stabilizer AME state exactly when all these party-aligned square block maps are isomorphisms. Moreover, $K_{I,J}$ is invertible if and only if the complementary block $K_{B \setminus I, B^c \setminus J}$ is invertible. Thus it suffices to test one representative of each complementary pair.

For each $j \notin B$, put $A_j = B \cup \{j\}$. The inverse image of the j-th coordinate summand is $L(A_j)$, and therefore

$$L = \bigoplus_{j \notin B} L(A_j).$$

Thus these m minimum-support subgroups determine the full stabilizer label group. Moreover, no collection of fewer than m minimum-support subgroups can span L.

Proof. The kernel of pr_{B^c} is $L(B) = \{0\}$ by the AME support condition. Its domain and codomain both have order q^{2m}, so it is an isomorphism. This gives the graph map K_B and the systematic check-matrix form. Since L is isotropic, evaluating the ambient direct-sum symplectic form on $(K_B x, x)$ and $(K_B y, y)$ gives the displayed identity.

For the block statement, let $S = I \cup (B^c \setminus J)$, so $|S| = m$. Projection $L \to \bigoplus_{s \in S} P_s$ is an isomorphism by the same kernel-and-order argument. If x is supported on J and $K_{I,J} x = 0$, then $(K_B x, x) \in L$ projects to zero on S. Thus $x = 0$. The source and target of $K_{I,J}$ have equal dimension, so the block map is an isomorphism.

Conversely, every m-party set has the form $I \cup (B^c \setminus J)$ with $|I| = |J|$. The kernel of projection from graph(K_B) to this set is naturally $\ker K_{I,J}$. If all the displayed block maps are isomorphisms, every half-set projection is therefore injective and hence bijective. Equivalently, there is no nonzero stabilizer label supported on at most m parties, which is the stabilizer AME condition.

For complementarity, $K_{I,J}$ is invertible exactly when $K_B(\bigoplus_{j \in J} P_j)$ is transverse to $\bigoplus_{i \in B \setminus I} P_i$. Taking symplectic orthogonal complements and using the anti-symplectic identity for K_B gives the same condition for $K_{B \setminus I, B^c \setminus J}$.

The inverse image of the j-th coordinate summand consists exactly of the labels supported on $B \cup \{j\}$. Pulling back the coordinate direct sum gives the displayed decomposition.

Every minimum-support subgroup has order q^2 by Proposition 3.4. A sum of r such subgroups has order at most q^{2r}, whereas $|L| = q^{2m}$. Spanning L therefore requires $r \geq m$. $\square$

We now prove the transition-map classification stated in the introduction. Regard each local label group

$$P_i = \mathbb{F}_q^2$$

as a $2e$-dimensional trace-symplectic space over $\mathbb{F}_p$. For every $(m+1)$-set A and $i, j \in A$, the support theorem defines the operator-pushing transition

$$M_A(i, j) = p_j p_i^{-1} : P_i \longrightarrow P_j.$$

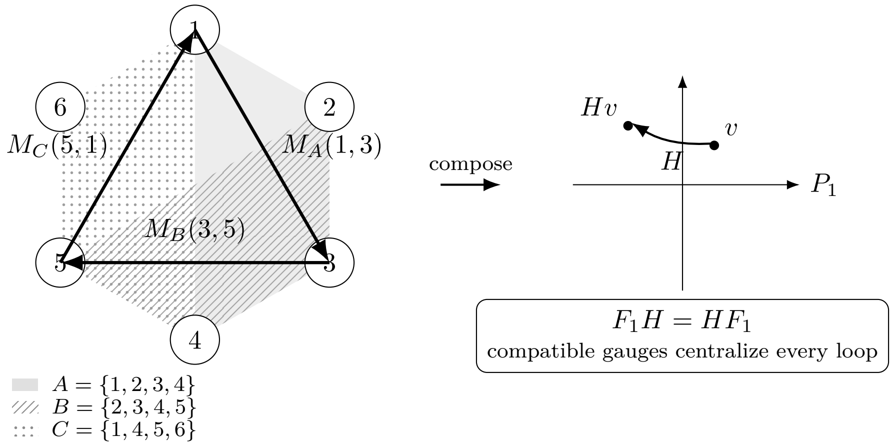

Figure 2: Operator pushing on minimum supports, shown for six parties. Each $(m+1)$-party support identifies the local Weyl-label planes of its parties through the bijective projections of the support theorem; overlapping supports compose to a loop whose holonomy $H = M_C M_B M_A$ is a linear map of the base plane, carrying a label v of the base plane to Hv. A local frame change is compatible precisely when it intertwines every transition; for one state over a prime field, its base block centralizes every loop holonomy.

For two states ψ, ϕ on the same labelled parties, a *symplectic equivalence of the transition systems* is a tuple $F_i \in \mathrm{Sp}(P_i)$ satisfying

$$F_j M_A^{\psi}(i,j) = M_A^{\phi}(i,j) F_i \quad \text{for every } A \text{ and } i,j \in A. \tag{4.2}$$

Here $\mathrm{Sp}(P_i) = \mathrm{Sp}_{2e}(\mathbb{F}_p)$ is the action induced by the standard additive local Clifford group.

For one state, define

$$\Gamma_\psi = \{[U] \in \mathrm{PU}(q^{2m}) : U = U_1 \otimes \cdots \otimes U_{2m},\ U|\psi\rangle \in \mathbb{C}|\psi\rangle\},$$
$$\mathrm{Aut}_{\mathrm{tr}}(\psi) = \{(F_i) \in \prod_i \mathrm{Sp}(P_i) : F_j M_A(i,j) = M_A(i,j) F_i \text{ for all } A, i, j \in A\}.$$

Here the subscript tr refers to the transition system; this group is also the linear image of the fixed-party transversal symmetries. Theorem 1.1 and the Clifford sequence (2.1) give a well-defined homomorphism

$$\lambda_\psi : \Gamma_\psi \longrightarrow \mathrm{Aut}_{\mathrm{tr}}(\psi), \qquad [U_1 \otimes \cdots \otimes U_{2m}] \longmapsto (\overline{U}_i)_i,$$

where $\overline{U}_i$ is the induced symplectic label map. A product decomposition is unique up to one-site scalars, which do not change $\overline{U}_i$. The injection $L_\psi \hookrightarrow \Gamma_\psi$ sends a label ℓ to the projective Pauli $[W_\ell]$.

Lemma 4.2 (Pauli phase correction)**.** Let ψ and ϕ be stabilizer states with Lagrangian label groups L_ψ and L_ϕ. If a product Clifford U induces a symplectic label map carrying L_ψ onto L_ϕ, then there is a product Pauli W_v such that $W_v U|\psi\rangle \in \mathbb{C}|\phi\rangle$. If L_ϕ is the label space of a stabilizer $\mathrm{AME}(2m,q)$ state, then for any prescribed m-party set C, the correcting label v may be chosen uniquely with support contained in C. No prescribed set of fewer than m parties can support corrections for every possible character discrepancy.

Proof. The state $U|\psi\rangle$ is stabilized on the same Pauli axes as $|\phi\rangle$, possibly with a different stabilizer character. The trace-symplectic pairing identifies the ambient label space modulo $L_\phi = L_\phi^\perp$ with the character dual of L_ϕ. Choose v whose pairing character is the inverse of the discrepancy. Conjugation by the product Pauli W_v cancels it. The corrected state and $|\phi\rangle$ therefore lie in the same one-dimensional joint stabilizer eigenspace.

For the final statement, projection $L_\phi \to \bigoplus_{i\in C} P_i$ is an isomorphism by Proposition 4.1. The trace-symplectic form on the latter space is nondegenerate. Hence pairing with labels supported on C identifies $\bigoplus_{i\in C} P_i$ with the full character dual of L_ϕ. Exactly one such label realizes the inverse discrepancy. If a prescribed support has $r < m$ parties, its label space has only q^{2r} elements, whereas the character group of L_ϕ has q^{2m}. It therefore cannot realize every discrepancy. □

Proof of Theorem 1.4. First fix the party labels. A block symplectic map $F = \bigoplus_i F_i$ carrying L_ψ to L_ϕ carries every supported subgroup $L_\psi(A)$ to $L_\phi(A)$. Writing these subgroups as graphs through any projection p_i gives exactly (4.2).

Conversely, suppose (4.2) holds. Fix A and $i \in A$. Every element of $L_\psi(A)$ is uniquely determined by its i-coordinate, and the equations with $j \in A$ show that its image under F is the unique element of $L_\phi(A)$ with i-coordinate $F_i p_i(\ell)$. Thus $FL_\psi(A) = L_\phi(A)$. Since the $(m+1)$-supported subgroups generate the full label groups by Proposition 4.1, $FL_\psi = L_\phi$.

Sequence (2.1) supplies local Clifford lifts of the F_i, and Lemma 4.2 supplies the product-Pauli correction. Hence an equivalence of transition systems lifts to an LC equivalence. The reverse implication follows from the induced label action, and Theorem 1.1 replaces LC by LU. A party permutation is absorbed by relabelling first.

For automorphisms, λ_ψ is surjective by the same lifting and phase-correction argument. Its kernel consists of projective product Paulis by (2.1). Such a Pauli fixes the state ray exactly when its label lies in $L_\psi^\perp = L_\psi$. Thus the displayed injection of L_ψ identifies it with $\ker \lambda_\psi$, proving exactness.

□

Corollary 4.3 (Finite recognition bound)**.** Let ψ and ϕ be stabilizer $\mathrm{AME}(2m, q)$ states, $q = p^e$ and $m \geq 2$, given by check matrices for their additive label spaces. The AME property is an input promise, not a condition verified by this algorithm. With the party labels fixed, choose any m-party set B, and write their half-set systematic maps as K_B^ψ, K_B^ϕ. The states are LU equivalent if and only if there are $F_i \in \mathrm{Sp}_{2e}(\mathbb{F}_p)$ satisfying

$$\left(\bigoplus_{i\in B} F_i\right) K_B^\psi = K_B^\phi \left(\bigoplus_{j\notin B} F_j\right). \tag{4.3}$$

This can be decided by testing at most

$$|\operatorname{Sp}_{2e}(\mathbb{F}_p)| = p^{e^2} \prod_{r=1}^{e} (p^{2r} - 1)$$

base symplectic frames. In the arithmetic-operation model over $\mathbb{F}_p$, each candidate requires polynomial work in the check-matrix size. The output is either inequivalence or the compatible symplectic frames $(F_i)_i$. The stabilizer phases need not be supplied for this decision: Lemma 4.2 guarantees a character repair after the label spaces are matched. If the two characters are supplied, then for any chosen m-party set the repair is the unique Pauli label on that carrier realizing their discrepancy, and it is found by one linear solve. For prime q, the number of base candidates is $|\mathrm{SL}_2(q)| = q(q^2 - 1)$. In particular, fixed-label recognition is polynomial in m for fixed local dimension q. With classical cubic matrix arithmetic, a concrete bound is

$$O\big((me)^3 + |\operatorname{Sp}_{2e}(\mathbb{F}_p)|\, m^2 e^3\big)$$

$\mathbb{F}_p$-operations.

If the party relabelling is unknown, exhaustive search introduces at most a factor $(2m)!$. For one state,

$$|\Gamma_\psi| = q^{2m} |\operatorname{Aut}_{\mathrm{tr}}(\psi)| \le q^{2m} |\operatorname{Sp}_{2e}(\mathbb{F}_p)|.$$

Proof. Mapping the two graph subspaces into one another is exactly the block equation (4.3). By Theorem 1.4 and Lemma 4.2, its locally symplectic solutions are exactly the LU equivalences.

Write K^ψ_{ij} and K^ϕ_{ij} for the party blocks. They are invertible by Proposition 4.1. Choose $i_0 \in B$ and $j_0 \notin B$. Once F_{j_0} is fixed, the (i, j_0)-blocks of (4.3) determine every F_i, and the (i_0, j)-blocks then determine every remaining F_j. Enumerate $F_{j_0} \in \operatorname{Sp}_{2e}(\mathbb{F}_p)$, test that the propagated frames are symplectic, and check all m^2 block equations. Forming the systematic maps and carrying out these checks is finite-field linear algebra polynomial in the input size and gives the displayed arithmetic bound. The $2m-1$ propagation equations form a spanning tree in the complete bipartite party graph; the remaining $m^2 - (2m-1) = (m-1)^2$ checks are its fundamental four-cycles. Enumerating party permutations gives the stated factor. The group-order bound follows from the exact sequence in Theorem 1.4 and injectivity of evaluation at the base frame. ☐

Corollary 4.4 (Prime-field recognition by fundamental cycles)**.** Let ψ, ϕ be stabilizer $\mathrm{AME}(2m, q)$ states with $m \ge 2$, assume $q = p$, and fix a labelled split $B \sqcup B^c$. Write $G = K^\psi_B$, $H = K^\phi_B$, and choose $b \in B$ and $c \in B^c$. For $i \ne b$ and $j \ne c$, define

$$Q^G_{ij} = G_{bj} G^{-1}_{ij} G_{ic} G^{-1}_{bc} \in \mathrm{GL}_2(q),$$

and define Q^H_{ij} similarly. Then ψ and ϕ are fixed-party LU equivalent if and only if

$$\det G_{ij} = \det H_{ij} \quad \text{for all } i, j$$

and there is an $A \in \mathrm{SL}_2(q)$ such that

$$A Q^G_{ij} = Q^H_{ij} A \quad \text{for all } i \ne b,\ j \ne c.$$

Thus prime-field recognition reduces to homogeneous linear equations in the four entries of A, followed by the single quadratic condition $\det A = 1$. Equivalently, the block-determinant array together with the simultaneous $\mathrm{SL}_2(q)$-conjugacy class of the ordered fundamental holonomies is a complete fixed-party LU invariant. The corresponding symmetry algebra and its five prime-field types are identified intrinsically in Section 5.

Proof. Suppose first that

$$A_i G_{ij} = H_{ij} D_j$$

for local $A_i, D_j \in \mathrm{SL}_2(q)$. Taking determinants gives the block-determinant equalities. Multiplying around the displayed four-cycle gives $Q^H_{ij} = A_b Q^G_{ij} A_b^{-1}$, so $A = A_b$ satisfies the intertwining equations.

Conversely, start from such an A and set

$$D_j = H^{-1}_{bj} A G_{bj}, \qquad A_i = H_{ic} D_c G^{-1}_{ic}.$$

The determinant equalities put every D_j and A_i in $\mathrm{SL}_2(q)$. The base row and base column equations hold by construction. Each remaining equation $A_i G_{ij} = H_{ij} D_j$ is equivalent, after substitution, to $A Q^G_{ij} = Q^H_{ij} A$. Hence the systematic matrices are locally symplectically equivalent, and Corollary 4.3 lifts this to LU equivalence. Taking $G = H$ gives the centralizer statement. ☐

4.1 A four-qutrit example

The state

$$|\psi\rangle = \frac{1}{3} \sum_{x,y\in\mathbb{F}_3} |x, y, x+y, x+2y\rangle$$

connects the support, transition, and character constructions. The four coordinate forms have coefficient columns $(1,0),(0,1),(1,1),(1,2)$; every pair is independent over $\mathbb{F}_3$. Thus any two coordinates determine (x,y). Tracing out any two parties kills all off-diagonal terms and leaves each pair value once, giving $\rho_A = I/9$ for $|A| = 2$.

For vectors $a, b \in \mathbb{F}_3^4$, abbreviate $X(a) = \bigotimes_i X(a_i)$ and similarly for $Z(b)$. Four independent commuting stabilizer generators, all with eigenvalue one, are

$$X(1,0,1,1), \quad X(0,1,1,2), \quad Z(2,2,1,0), \quad Z(2,1,0,1).$$

The two X labels generate the computational-support code; the two Z labels generate its orthogonal complement. Hence a general label has

$$a = (x, y, x+y, x+2y), \qquad b = (-u-v, -u-2v, u, v).$$

Choose $B = \{1,2\}$ and write local labels as columns $v_i = (a_i, b_i)$. Solving for the first two parties gives the systematic graph map

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = K_B \begin{pmatrix} v_3 \\ v_4 \end{pmatrix}, \qquad K_B = \begin{pmatrix} 2I_2 & 2I_2 \\ 2I_2 & I_2 \end{pmatrix}.$$

The row-generator check matrix is therefore $[K_B^{\mathsf{T}} \mid I_4]$ in the party order $1,2,3,4$. Its two coordinate summands are

$$L(\{1,2,3\}) = \{(2w, 2w, w, 0) : w \in \mathbb{F}_3^2\}, \qquad L(\{1,2,4\}) = \{(2w, w, 0, w) : w \in \mathbb{F}_3^2\}.$$

They intersect trivially and span L. The first subgroup is generated by $X(1,1,2,0)$ and $Z(2,2,1,0)$. In particular, $M_{\{1,2,3\}}(3,1) = 2I_2$: one local Weyl label fixes the other two.

For the base row $b = 1$ and column $c = 3$, there is just one fundamental cycle. With $G = K_B$, it is

$$Q_{24} = G_{14}G_{24}^{-1}G_{23}G_{13}^{-1} = (2I_2)I_2(2I_2)(2I_2) = 2I_2.$$

Thus every $A \in \mathrm{SL}_2(3)$ commutes with the cycle; propagation gives the same frame A at all four sites. This illustrates how one base frame determines the compatible tuple. Here the cycle is scalar, so it imposes no further restriction on the base frame; nonscalar cycles constrain that choice in the general recognition problem.

Finally let $|\phi\rangle = Z_1(1)|\psi\rangle$ and $\omega = e^{2\pi i/3}$. Its label space and transition maps are unchanged, but the first displayed X generator has eigenvalue ω^{-1} on ϕ; the other three still have eigenvalue one. The identity symplectic witness alone therefore does not identify the state ray. The correction $Z_1(-1)$ carries ϕ to ψ, and is the unique correcting label supported on the prescribed half-set $\{1,2\}$.

There is also an elementary extension-field variant. Replace $\mathbb{F}_3$ by $\mathbb{F}_9$ and the normalization by $1/9$. The same pairwise independence proves AME. The basis permutation $C|x\rangle = |x^3\rangle$ satisfies

$$CX(a)C^\dagger = X(a^3), \qquad CZ(b)C^\dagger = Z(b^3),$$

because the field trace is invariant under Frobenius. Applying C at all four sites permutes the summands and fixes the state. The label map $(a,b) \mapsto (a^3, b^3)$ is trace-symplectic over $\mathbb{F}_3$, but is not $\mathbb{F}_9$-linear. This actual local symmetry belongs to $\mathrm{Sp}_4(\mathbb{F}_3)$ and would be lost by restricting to $\mathrm{SL}_2(\mathbb{F}_9)$.

4.2 Determination from marginals

The support constraints determine the state among all density operators and give a parent Hamiltonian. We first prove exact determination, then ask how reliably the constraints detect an imperfect preparation. For a party set A, write σ_A for the reduction of a density operator σ to A, and $\|X\|_1 = \operatorname{Tr}|X|$. An operator on a party set is identified with its tensor product by the identity on the remaining parties.

Proposition 4.5 (Marginal certification)**.** Let $|\psi\rangle$ be a stabilizer AME$(2m,q)$ state with $m \geq 2$, fix an m-party set B, and for $j \notin B$ put $A_j = B \cup \{j\}$ and

$$\Pi_j = q^{m-1}\rho^{\psi}_{A_j} \otimes I_{A_j^c}.$$

1. Each Π_j is the orthogonal projector onto the joint $+1$ eigenspace of the stabilizer lifts with labels in $L(A_j)$. The Π_j commute, and $\prod_{j\notin B}\Pi_j = |\psi\rangle\langle\psi|$.
2. A density operator σ on the $2m$ parties with $\sigma_{A_j} = \rho^{\psi}_{A_j}$ for every $j \notin B$ equals $|\psi\rangle\langle\psi|$. More generally, the reductions of ψ to $(m+1)$-party sets $C_1, \dots, C_k$ determine ψ among all density operators if and only if $L(C_1), \dots, L(C_k)$ span L; in particular $k \geq m$.
3. Every reduction of ψ to at most m parties is maximally mixed, so no family of such reductions distinguishes ψ from the maximally mixed state.
4. $H = \sum_{j\notin B}(I - \Pi_j)$ is a sum of m commuting projectors, each acting on $m+1$ parties, with kernel $\mathbb{C}|\psi\rangle$ and spectral gap exactly one. For every density operator σ,

$$1 - \langle\psi|\sigma|\psi\rangle \leq \operatorname{Tr}(H\sigma) \leq \sum_{j\notin B} \tfrac{1}{2}\big\|\sigma_{A_j} - \rho^{\psi}_{A_j}\big\|_1.$$

5. A Hermitian operator that is a sum of terms acting on at most m parties each and has $|\psi\rangle$ as a ground state is a scalar multiple of the identity.

Proof. (i) By (2.3) and Proposition 3.4,

$$q^{m-1}\rho^{\psi}_{A_j} \otimes I_{A_j^c} = q^{-2} \sum_{\ell\in L(A_j)} \widetilde{W}_\ell,$$

and the lifts on the right form the subgroup of the stabilizer group with labels in $L(A_j)$, of order q^2. The normalized sum over a finite abelian group of unitaries is the projector onto its joint $+1$ eigenspace, so Π_j is that projector, and the Π_j commute because the stabilizer group is abelian. By Proposition 4.1, addition is a bijection $\bigoplus_{j\notin B} L(A_j) \to L$, and the lift is multiplicative on the stabilizer group; hence

$$\prod_{j\notin B}\Pi_j = q^{-2m}\sum_{\ell\in L}\widetilde{W}_\ell = |\psi\rangle\langle\psi|$$

by (2.2).

(ii) Let $G = L(C_1) + \cdots + L(C_k)$ and $\Pi_{C_i} = q^{-2}\sum_{\ell\in L(C_i)}\widetilde{W}_\ell$, a projector by the argument of (i). If $\sigma_{C_i} = \rho^{\psi}_{C_i}$, then Weyl orthogonality and $|L(C_i)| = q^2$ give $\operatorname{Tr}(\sigma\Pi_{C_i}) = q^{m-1}\operatorname{Tr}\big((\rho^{\psi}_{C_i})^2\big) = 1$; since $0 \leq \Pi_{C_i} \leq I$, this forces $\Pi_{C_i}\sigma = \sigma$. If the $L(C_i)$ span L, then $\prod_i \Pi_{C_i}$ is the projector onto the joint $+1$ eigenspace of the whole stabilizer group, which is $|\psi\rangle\langle\psi|$, so $\sigma = |\psi\rangle\langle\psi|\sigma$; taking adjoints and using $\operatorname{Tr}\sigma = 1$, $\sigma = |\psi\rangle\langle\psi|\sigma|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi|$. If instead $G \neq L$, put

$$\tau = q^{-2m}\sum_{\ell\in G}\widetilde{W}_\ell.$$

This is $|G|q^{-2m}$ times the projector onto the joint $+1$ eigenspace of the lifts of G, whose rank $q^{2m}/|G|$ exceeds one; so τ is a density operator of rank greater than one. Tracing out C_i^c annihilates every lift not supported in C_i, and $G \cap L(C_i) = L(C_i)$, so

$$\tau_{C_i} = q^{-2m} \cdot q^{m-1} \sum_{\ell \in L(C_i)} \widetilde{W}_\ell = \rho^\psi_{C_i}$$

by (2.3). Thus the reductions do not determine ψ. The first sentence is the case $C_i = A_j$, where Proposition 4.1 gives the spanning, and its final sentence gives $k \geq m$.

(iii) is (2.3) with the AME condition.

(iv) Commuting projectors make H diagonalizable with integer eigenvalues, and the eigenvalue 0 has eigenspace $\bigcap_j \operatorname{range} \Pi_j = \operatorname{range} \prod_j \Pi_j = \mathbb{C}|\psi\rangle$ by (i). Hence $H \geq I - |\psi\rangle\langle\psi|$, which is the first inequality. The gap is attained. Since $L^\perp = L$, pairing with labels $v \in (\mathbb{F}_q^2)^{2m}$ realizes every character of L; choose v whose character is nontrivial on $L(A_{j_0})$ and trivial on the other summands. Then $\Pi_{j_0} W_v|\psi\rangle = 0$ and $\Pi_j W_v|\psi\rangle = W_v|\psi\rangle$ for $j \neq j_0$, so $W_v|\psi\rangle$ is an eigenvector of H with eigenvalue one. For the second inequality, $q^{m-1}\rho^\psi_{A_j}$ is a projector on A_j by (i) and $\operatorname{Tr}\big(\rho^\psi_{A_j} \cdot q^{m-1}\rho^\psi_{A_j}\big) = 1$, so

$$\operatorname{Tr}\big(\sigma(I - \Pi_j)\big) = \operatorname{Tr}\big((\rho^\psi_{A_j} - \sigma_{A_j})\, q^{m-1}\rho^\psi_{A_j}\big).$$

For a traceless Hermitian X and $0 \leq P \leq I$, $\operatorname{Tr}(XP) \leq \operatorname{Tr} X_+ = \frac{1}{2}\|X\|_1$. Sum over j.

(v) Let $H' = \sum_k h_k$ with h_k acting on a set S_k of at most m parties. By (iii), $\langle\psi|h_k|\psi\rangle = q^{-|S_k|} \operatorname{Tr}_{S_k} h_k$, which is q^{-2m} times the trace of h_k on the full space; hence $\langle\psi|H'|\psi\rangle = q^{-2m} \operatorname{Tr} H'$ is the mean of the eigenvalues of H'. If ψ is a ground state, the least eigenvalue equals the mean, so all eigenvalues coincide. □

For qubit stabilizer states, Wu, Yang, Wang, Wen, Qin, and Gao show that the reductions to the supports of any n generators determine the state among all density operators [WYW+15, Theorem 2]. The AME support geometry lowers the count from $2m$ to m, fixes every support to $m+1$ parties, and shows that neither number can be reduced: no family of reductions to at most m parties determines ψ at all, and at support size $m+1$ fewer than m reductions never suffice. Generic pure states on n parties are determined among pure states by their reductions to about $n/2+1$ parties [JL05]; here the locality is the same, the determination is among all density operators, and the family is explicit. The spanning criterion in (ii) uses only (2.2)–(2.3) and the projector identity, so it holds for every stabilizer state: reductions to a family of party sets determine the state among all density operators exactly when the supported label subgroups span its label group. The certificate in (iv) is the stabilizer witness of [TG05] applied to a frustration-free Hamiltonian, and the state σ need not be pure, stabilizer, or AME, which separates it from the approximate product-symmetry statements of Section 6.

4.3 Testing the marginal constraints

Determination is an exact statement. To use the marginals as a certificate, we also need to know how reliably they detect a deviation from the target. Sampling the m projectors above uniformly gives a worst-case rejection rate of only $1/m$ on states orthogonal to ψ. We can improve this by using the complementary half as well.

Fix complementary halves B, C, and put

$$\mathcal{S} = \{B \cup \{j\} : j \in C\} \cup \{C \cup \{i\} : i \in B\}, \qquad \Pi_S = q^{m-1}\rho^\psi_S \otimes I_{S^c}.$$

In one round, choose S uniformly from these $2m$ supports and measure the binary test $\{\Pi_S, I - \Pi_S\}$, accepting on Π_S. This requires access to $m+1$ parties of one copy. Joint operations within that subset are allowed. The restriction is on the *total support* of a test; it differs from measuring all parties separately and combining their outcomes. A binary test need not be one laboratory measurement basis.

The verification-operator and spectral-gap framework is standard [PLM18, DHZ20], as are fidelity certificates from stabilizer generators and frustration-free Hamiltonians [KKL19, ZLC24]. Setting minimization and qudit stabilizer witnesses have also been studied [LZLZ21, SMA26]. Here the AME support geometry lets us calculate the optimal rejection rate under the stated support restriction.

Proposition 4.6 (Complementary marginal tests)**.** Let ψ be a stabilizer AME$(2m, q)$ state with $m \geq 2$. The verification operator and its gap are

$$\Omega_* = \frac{1}{2m} \sum_{S \in \mathcal{S}} \Pi_S, \qquad \nu_* = \frac{m+1}{2m}, \qquad I - \Omega_* \geq \nu_*(I - |\psi\rangle\langle\psi|). \tag{4.1}$$

The coefficient ν_* is optimal among randomized perfectly complete binary tests, each supported on at most $m+1$ parties. For every density operator σ, its rejection probability $r(\sigma) = \mathrm{Tr}[(I - \Omega_*)\sigma]$ satisfies

$$1 - \langle\psi|\sigma|\psi\rangle \leq \frac{r(\sigma)}{\nu_*} \leq \frac{1}{m+1} \sum_{S \in \mathcal{S}} \frac{1}{2} \|\sigma_S - \rho_S^{\psi}\|_1. \tag{4.2}$$

Proof. The key is that *any* m of the supported spaces $L(S)$ form a direct sum equal to L. To see this, select $I \subset B, J \subset C$ with $|I| + |J| = m$. In a relation $\sum_{i \in I} u_i = -\sum_{j \in J} v_j$, with $u_i \in L(C \cup \{i\})$ and $v_j \in L(B \cup \{j\})$, the common sum is supported in $I \cup J$. The support theorem makes it zero. Projection to i isolates u_i and is injective on its subgroup, so every term vanishes; likewise for v_j. Each subgroup has dimension $2e$ over $\mathbb{F}_p$, so the dimensions add to $\dim L$.

Use the actual stabilizer lifts fixing ψ. Their joint eigenbasis is indexed by characters of L. A projector Π_S accepts exactly those characters trivial on $L(S)$. A nontrivial character cannot pass m tests, hence fails at least $m+1$. This proves the operator inequality. It is sharp: any $m-1$ chosen subgroups have a nonzero annihilating character, which passes exactly those tests.

For optimality in the larger class of allowed tests, a perfectly complete effect E satisfies $E\psi = \psi$. A traceless one-site unitary V_i makes $V_i\psi$ orthogonal to ψ, and every test omitting i still accepts it. Thus the gap is at most the probability of including party i. Averaging over $2m$ parties bounds that probability, for some i, by $(m+1)/(2m)$. Finally take the expectation of the operator inequality. Testing a marginal discrepancy against its support projector bounds each rejection probability by $\frac{1}{2}\|\sigma_S - \rho_S^{\psi}\|_1$; average and divide by ν_*. □

For the four-qutrit state of Section 4.1, the two tests on 123 and 124 have gap 1/2; using all four three-party tests raises it to 3/4. The state $Z_1(1)\psi$ passes the test omitting party 1 and fails the other three: on each support containing party 1, surjectivity onto its Weyl labels makes the induced character nontrivial. These tests detect exactly the character change that left the transition maps unchanged in the example.

Only acceptance probabilities are needed for the first fidelity bound; full marginal tomography is unnecessary. If the prepared state is promised to be $U\psi$ for a product unitary U, the same certificate enters the robust rigidity theorem through Corollary 6.11. Appendix C gives the exact tradeoff for fewer or unequally weighted tests and extends the uniform gap to every AME state.

4.4 Stochastic conversion

The next consequence removes the unitarity assumption from exact rigidity.

Corollary 4.7 (Stochastic conversion)**.** Let $|\psi\rangle, |\phi\rangle$ be stabilizer AME$(2m, q)$ states with $m \geq 2$, let π be a party permutation, and let $A_1, \ldots, A_{2m}$ be linear operators on $\mathbb{C}^q$ with

$$(A_1 \otimes \cdots \otimes A_{2m})\pi|\psi\rangle = c|\phi\rangle, \qquad c \neq 0.$$

Then every A_i is a nonzero scalar multiple of a Clifford unitary. Consequently, on the set of stabilizer AME$(2m, q)$ states with $m \geq 2$, the relations of equivalence under stochastic local operations and classical communication, LU equivalence, and LC equivalence coincide; the optimal probability of exact single-copy conversion is one if the states are LC equivalent and zero otherwise; and Corollary 4.3 decides which, with an unknown party relabelling handled as there.

Proof. Replacing ψ by the stabilizer AME state $\pi|\psi\rangle$, we may take $\pi = 1$. For each i, $|c|^2\rho_i^\phi = A_i M A_i^\dagger$ with $M \geq 0$, and $\rho_i^\phi = q^{-1}I$ has rank q, so each A_i is invertible. Write $A_i = c_i U_i e^{H_i}$ with $c_i \neq 0$, U_i unitary, and H_i Hermitian and traceless: this is the polar decomposition, with the scalar chosen to remove the trace of the logarithm of the positive factor. Let $H = \sum_i H_i$ act on party i through H_i, and put

$$f(t) = \log\bigl\|e^{tH}|\psi\rangle\bigr\|^2, \qquad |\psi_t\rangle = e^{tH}|\psi\rangle/\bigl\|e^{tH}|\psi\rangle\bigr\|.$$

Then $f'(t) = 2\langle\psi_t|H|\psi_t\rangle$ and $f''(t) = 4\,\mathrm{Var}_{\psi_t}(H) \geq 0$. The states $\psi_0 = \psi$ and $\psi_1 \propto (\bigotimes_i U_i^\dagger)\phi$ both have one-party reductions $q^{-1}I$, and each H_i is traceless, so $f'(0) = f'(1) = 0$. Convexity forces $f' \equiv 0$ on $[0, 1]$, and f is real analytic, so $f''(0) = 0$, that is $\mathrm{Var}_\psi(H) = 0$ and $H|\psi\rangle = 0$. Expanding $\langle\psi|H^2|\psi\rangle = 0$ with the two-party reductions $q^{-2}I$ of ψ, which is where $m \geq 2$ enters, gives

$$0 = \sum_i \langle\psi|H_i^2|\psi\rangle + \sum_{i \neq j} \langle\psi|H_i H_j|\psi\rangle = q^{-1}\sum_i \mathrm{Tr}(H_i^2),$$

so every $H_i = 0$ and $A_i = c_i U_i$. Theorem 1.1 makes each U_i Clifford. A successful branch of a single-copy stochastic protocol is a product of Kraus operators as displayed, with local output spaces $\mathbb{C}^q$ forced by the target, so no branch of any protocol reaches an LC-inequivalent state, while LC-equivalent states are converted deterministically. □

Two states whose one-party reductions are all maximally mixed are SLOCC equivalent if and only if they are LU equivalent, by the Kempf–Ness theorem [KN79, GW11]; this is recorded for AME states in [BR20, Corollary 1], and the resulting conversion dichotomy in [RMBR+26]. Kempf–Ness alone does not make each factor unitary: the stabilizer of such a state in the product of special linear groups is reductive and may contain a torus, as it does for the Greenberger–Horne–Zeilinger state, which is one-uniform. The variance argument uses two-uniformity and applies verbatim to every two-uniform state; the Clifford conclusion is the stabilizer-AME rigidity. For qubit stabilizer states, compare the local-symmetry classification of [EK20].

4.5 Encoder consequences

Proof of Corollary 1.3. The input marginal of each AME Choi state is $q^{-1}I$, so one-leg reshaping gives an isometry. The projection of its stabilizer label group onto the input Weyl-label space is surjective: otherwise a nonzero local label orthogonal to the image would lie in the maximal isotropic space $L^\perp = L$, contradicting the absence of a one-party stabilizer. The kernel therefore has q^{2m-2} elements and stabilizes the image, making it a one-logical-qudit stabilizer code. AME corrects erasure of any $m-1$ outputs, so its distance is at least m; the quantum Singleton bound gives equality. This is the standard perfect-tensor/quantum-MDS correspondence used in [Tan26b, Theorems 3.5 and 3.6].

These particular encodings are Pauli-compatible. For a stabilizer lift $A \otimes T$ of the Choi state, vectorization gives $TV = V(A^{\mathsf{T}})^{-1}$. Surjectivity onto the reference labels therefore supplies a physical Pauli representative of every input Weyl, with its scalar adjusted to give equality. This property concerns the specified encoder, not just its image: replacing V by VR for an arbitrary logical unitary R need not preserve it.

Let $|\Phi\rangle = q^{-1/2}\sum_x |x,x\rangle$, and use the Choi convention $|\psi\rangle = (I \otimes V_\psi)|\Phi\rangle$ and $|\phi\rangle = (I \otimes V_\phi)|\Phi\rangle$. Writing $U_{\text{phys}} = \bigotimes_i U_i$, unitarity and the isometry relation give $L^\dagger L = I$, hence $L \in U(q)$. The relation $U_{\text{phys}} V_\psi = V_\phi L$ gives

$$(I \otimes U_{\text{phys}})|\psi\rangle = (L^T \otimes I)|\phi\rangle.$$

Hence

$$\big((L^T)^{-1} \otimes U_{\text{phys}}\big)|\psi\rangle = |\phi\rangle.$$

Theorem 1.1 makes every displayed local factor Clifford. Entrywise conjugation preserves the finite-field Weyl group, since $\overline{X(a)} = X(a)$ and $\overline{Z(b)} = Z(-b)$; it therefore preserves the Clifford normalizer. Taking adjoints preserves the normalizer as well, and $U^T = (\overline{U})^\dagger$. Thus transposition preserves the finite-field Clifford group, and L is Clifford. $\square$

The recognition procedure is constructive for these encoders. Given check matrices for the two normalized Choi states, Corollary 4.3 decides whether a transversal conversion exists and returns compact symplectic-frame witnesses within the same arithmetic bound. If full stabilizer tableaux and chosen Clifford lifts are also supplied, one further linear solve returns a compact Clifford–Pauli conversion. If the reference Clifford lift is C with symplectic label map F, its logical unitary and label map are respectively

$$(C^{\mathsf{T}})^{-1} = \overline{C}, \qquad F_{\text{logical}} = \tau F \tau, \qquad \tau(a,b) = (a,-b).$$

Indeed $\overline{W_{(a,b)}} = W_{\tau(a,b)}$. The label formula is generally different from $F^{-\mathsf{T}}$. Lemma 4.2 allows the character repair to be supported on any prescribed m physical outputs. Thus the repair need not alter the logical factor. Moreover, m is the smallest size of a prescribed carrier that works uniformly for every discrepancy. This bound does not include expansion into dense unitary matrices.

5 The local endomorphism algebra

The preceding results solve LU equivalence for every prime power. This section extracts additional structure from the same systematic matrix. The algebraic description below is valid over the underlying prime field; the five-type classification and its code interpretations specialize explicitly to prime local dimension. None of this section is needed for the robust rounding theorem.

Let $m \geq 2$, and let $L \subseteq \bigoplus_i P_i$ be the label space of a stabilizer AME$(2m, q)$ state, where each P_i is regarded as a $2e$-dimensional $\mathbb{F}_p$-space. Define

$$\mathcal{E}_L = \left\{ (T_i)_i \in \prod_i \operatorname{End}_{\mathbb{F}_p}(P_i) : \left(\bigoplus_i T_i \right) L \subseteq L \right\}. \tag{5.1}$$

Thus $\mathcal{E}_L$ is the algebra of block-diagonal, one-party label endomorphisms preserving L.

Theorem 5.1 (Endomorphism algebra from fundamental cycles)**.** Fix a half-set $B \sqcup B^c$, write $G = K_B$, and choose $b \in B$ and $c \in B^c$. For $i \neq b$ and $j \neq c$, put

$$Q_{ij} = G_{bj} G_{ij}^{-1} G_{ic} G_{bc}^{-1} \in \mathrm{GL}_{2e}(\mathbb{F}_p).$$

Evaluation at party b is an algebra isomorphism

$$\mathcal{E}_L \;\cong\; \mathrm{Cent}_{\mathrm{End}_{\mathbb{F}_p}(P_b)}\{Q_{ij} : i \neq b,\ j \neq c\}. \tag{5.2}$$

Under this isomorphism, the compatible local symplectic automorphisms are the units whose propagated components preserve the symplectic form at every party. If $q = p$, this condition is simply

$$\mathrm{Aut}_{\mathrm{tr}}(\psi) \cong \{T \in \mathcal{E}_L : \det T_b = 1\}. \tag{5.3}$$

In particular, the abstract algebra does not depend on the chosen half-set or base parties.

Proof. A block-diagonal endomorphism, written as $(A_i)_{i \in B}$ and $(D_j)_{j \in B^c}$, preserves the graph of G exactly when

$$A_i G_{ij} = G_{ij} D_j \qquad (i \in B,\ j \in B^c). \tag{5.4}$$

Every block G_{ij} is invertible by Proposition 4.1. Once $A_b = A$ is fixed, the base row and column force

$$D_j = G_{bj}^{-1} A G_{bj}, \qquad A_i = G_{ic} D_c G_{ic}^{-1}. \tag{5.5}$$

The remaining equations are precisely $AQ_{ij} = Q_{ij}A$. Evaluation at b is therefore bijective onto the common centralizer, and (5.5) shows that it respects addition and multiplication.

An automorphism is compatible exactly when every propagated component in (5.5) is symplectic. For $q = p$, all components are conjugate 2×2 matrices and $\mathrm{Sp}_2(q) = \mathrm{SL}_2(q)$, so this is equivalent to $\det T_b = 1$. For $e > 1$, one must retain all transported symplectic-form conditions; a single determinant condition is not sufficient. Finally, the definition in (5.1) is independent of the systematic presentation. ☐

For prime dimension, Theorem 5.1 replaces the former list of unrelated group orders by a classification of one algebra. It also explains what additional code structure each nonscalar case supplies.

Corollary 5.2 (Five prime-field algebra types)**.** Assume $m \geq 2$ and $q = p$, and put $d = \gcd(2, q-1)$. Exactly one row of Table 1 applies. The middle two columns give $\mathrm{Aut}_{\mathrm{tr}}(\psi)$ and its order, and

$$|\Gamma_\psi| = q^{2m} |\mathrm{Aut}_{\mathrm{tr}}(\psi)|.$$

After compatible local coordinates are chosen, the last column gives the code or module structure forced by $\mathcal{E}_L$.

$\mathcal{E}_L$	compatible group	order	resulting structure
$M_2(\mathbb{F}_q)$	$\mathrm{SL}_2(q)$	$q(q^2-1)$	$\mathbb{F}_q^2 \otimes C$, where C is a weighted self-dual classical MDS code; an MDS–CSS stabilizer after local symplectic coordinate changes
$\mathbb{F}_q I$	center of $\mathrm{SL}_2(q)$	d	no additional linear structure is forced
$\mathbb{F}_q \times \mathbb{F}_q$	split norm-one torus	$q-1$	two weighted-dual classical MDS codes on complementary local axes; an MDS–CSS stabilizer after local symplectic coordinate changes
$\mathbb{F}_{q^2}$	field norm-one torus	$q+1$	an $\mathbb{F}_{q^2}$-linear MDS code, weighted Hermitian self-dual in compatible coordinates
$\mathbb{F}_q[\epsilon]/(\epsilon^2)$	determinant-one units	qd	a free rank-m dual-number module for which every half-set projection is an isomorphism

Table 1: The five prime-field endomorphism algebras, their determinant-one unit groups, and the code structures they force.

Proof of Corollary 5.2. If every fundamental holonomy is scalar, its common centralizer is $M_2(\mathbb{F}_q)$. Otherwise choose a nonscalar holonomy Q. Its centralizer is the quadratic algebra $\mathbb{F}_q[Q]$. If another holonomy does not commute with Q, the common centralizer is scalar; if all holonomies commute, the common centralizer remains $\mathbb{F}_q[Q]$. The three quadratic possibilities are the split algebra, the quadratic field, and the dual numbers. The determinant restricts respectively to the product, the field norm, and $\det(aI + bN) = a^2$ for $N^2 = 0$. This gives the five groups and orders in the table, including characteristic two.

The code interpretations follow from the action of $\mathcal{E}_L$ on the local planes and the fact that every half-set projection of L is an isomorphism. Appendix A gives the module argument. □

The classification is constructive without enumerating a finite group. Indeed, $\mathcal{E}_L$ is the kernel of the homogeneous linear system $[A, Q_{ij}] = 0$ in the four entries of A. Its dimension is 1, 2, or 4; in the middle case the characteristic polynomial of one nonscalar element distinguishes the three quadratic algebras. Propagation by (5.5) then recovers the one-party actions. Thus the recognition calculation also returns the symmetry type and the corresponding code-structure row of Table 1.

The constituent CSS and Hermitian code constructions are standard; see, for example, [KKKS06]. In the qubit case, invariance under a uniform order-three local Clifford was already identified with $\mathbb{F}_4$-linearity [VdNDDM05, Theorem 2]. Our use of the table is the converse organization specific to stabilizer AME states: the intrinsic endomorphism algebra determines both the compatible group and the code structure. We do not call the dual-number row an MDS code over a ring.

The algebra formulation also reveals a restriction at small party number that is absent from the five-row list.

Theorem 5.3 (Forced endomorphisms for four and six parties)**.** Let $q = p$. For every stabilizer AME$(2m, q)$ state with $m = 2$ or $m = 3$,

$$\dim_{\mathbb{F}_q} \mathcal{E}_L \geq 2. \tag{5.6}$$

Equivalently, all fundamental four-cycle holonomies commute, so the scalar row of Table 1 cannot occur. If $q \geq 5$, the state consequently has a noncentral compatible local symplectic automorphism.

Proof. For $m = 2$ there is only one fundamental holonomy, whose centralizer has dimension at least two.

Let $m = 3$. Independent invertible changes of block row and column bases put the 3×3 block matrix G in the form

$$N = \begin{pmatrix} I & I & I \\ I & A & B \\ I & C & D \end{pmatrix}. \tag{5.7}$$

These basis changes need not be symplectic. In dimension two, however, every invertible change rescales the alternating form. Pulling the domain and codomain forms through the block-diagonal basis changes therefore turns the anti-symplectic identity for G into the following weighted one. Explicitly, if $N = PGQ$, with $P = \operatorname{diag}(P_i)$ and $Q = \operatorname{diag}(Q_j)$, then

$$S^{\mathsf{T}} JS = (\det S)J \quad \Longrightarrow \quad P_i^{-\mathsf{T}} JP_i^{-1} = (\det P_i)^{-1} J, \qquad Q_j^{\mathsf{T}} JQ_j = (\det Q_j)J.$$

Substitution in $G^{\mathsf{T}} \operatorname{diag}(J)G = -\operatorname{diag}(J)$ gives

$$N^{\mathsf{T}} \operatorname{diag}(r_0 J, r_1 J, r_2 J)N = -\operatorname{diag}(s_0 J, s_1 J, s_2 J) \tag{5.8}$$

with $r_i = (\det P_i)^{-1}$ and $s_j = \det Q_j$. Applying the same observation to N^{-1} gives $t_j = s_j^{-1}$, so distinct block rows satisfy

$$\sum_{j=0}^{2} t_j N_{ij} J N_{kj}^{\mathsf{T}} = 0 \qquad (i \neq k). \tag{5.9}$$

Thus (5.8) gives weighted column orthogonality and (5.9) gives weighted row orthogonality.

Orthogonality of the first two block rows gives

$$t_0 I + t_1 A^{\mathsf{T}} + t_2 B^{\mathsf{T}} = 0,$$

so $B \in \operatorname{span}_{\mathbb{F}_q}\{I, A\}$. The first and third rows similarly give $D \in \operatorname{span}\{I, C\}$. Orthogonality of the first two block columns gives

$$r_0 I + r_1 A + r_2 C = 0,$$

so C, and hence D, also lies in $\operatorname{span}\{I, A\}$. All four non-tree blocks commute. In the normalization (5.7), the fundamental holonomies are their inverses, so they commute as well. The common centralizer is therefore at least two-dimensional.

For $q \geq 5$, every determinant-one group attached to a nonscalar row of Table 1 has order strictly larger than $d = \gcd(2, q-1)$, which proves the final assertion. At $q = 3$, for example, the split norm-one group can coincide with the center; this is why the qualification is necessary. □

Thus extra label-space linearity is forced through six parties, without an occurrence or genericity claim for the remaining types.

6 Cleaning-based quantitative rounding

We now turn from exact equivalence to approximate symmetry. Estimating one reduced operator would lose the exponentially small nonidentity signal of a minimum-support marginal. The global argument avoids that loss. Choose one AME party as a logical input. The other $2m-1$ parties form an $[[2m-1, 1, m]]_q$ stabilizer code, and the product of their local factors approximately implements the transpose-inverse of the factor on the chosen party. Three correctable regions then test that logical action by a nested commutator. Indeed, distance m makes every set of at most $m-1$ erased physical parties correctable, and the $2m-1$ physical parties split into regions of sizes $m-1, m-1, 1$.

The exact mechanism is standard: the qubit stabilizer cleaning lemma is due to Bravyi and Terhal [BT09, Lemma 1]; in the additive prime-power setting used here, the same cleaning step is the elementary symplectic statement recorded in the proof of Lemma 6.1. Nested-commutator localization underlies the Bravyi–König hierarchy theorem [BK13], and Pastawski–Yoshida prove directly that a transversal logical unitary on three cleanable regions is Clifford [PY15, Lemma 5]. Disjointness gives a complementary exact-code hierarchy bound [JOKY18, Theorem 5]. The point below is therefore not a new qualitative transversal-gate obstruction. It is the robust step: an implementation with leakage error ε forces every nested logical Weyl commutator within 8ε of a scalar, after which a Fourier argument gives an explicit Clifford distance.

Figure 3 gives a roadmap for the local estimates, the Clifford separation step, and the resulting explicit threshold.

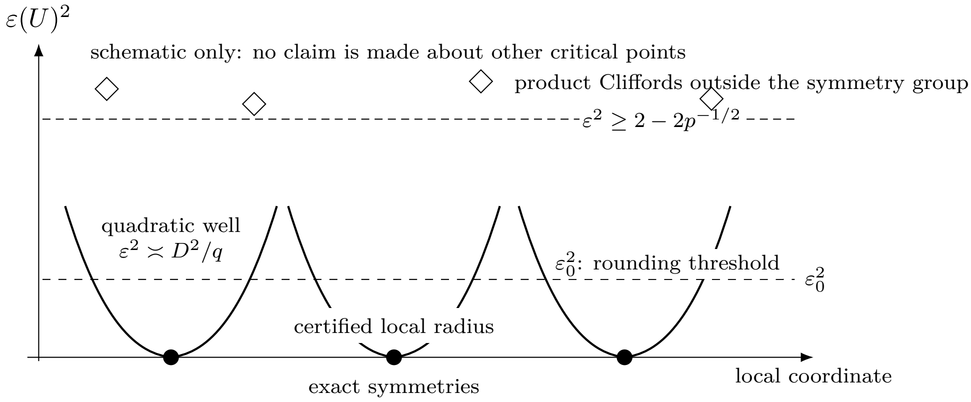

Figure 3: Roadmap for the defect estimates (a one-dimensional schematic of a high-dimensional projective product-unitary group). Modulo scalar phases, exact symmetries are isolated zeros with certified local quadratic growth. Every product Clifford outside the symmetry group has defect at least $(2-2p^{-1/2})^{1/2}$; below the explicit cleaning threshold of Section 6, an arbitrary product unitary rounds to a Clifford and then to an exact symmetry. The picture asserts neither that every local minimum is Clifford nor that the displayed coordinate exhausts the group.

For an encoding isometry $V : \mathbb{C}^q \to \mathcal{H}$, write

$$E(T, L) = q^{-1/2} \|TV - VL\|_{\mathrm{HS}}.$$

For physical unitaries T, S and logical unitaries L, M, it satisfies $E(TS, LM) \leq E(T, L) + E(S, M)$ and $E(T^\dagger, L^\dagger) = E(T, L)$.

Lemma 6.1 (Quantitative cleaning commutator)**.** Let V be a Pauli-compatible encoding of one logical qudit in a stabilizer code: every input Weyl P has a physical Pauli representative $\widetilde{P}$ with $\widetilde{P}V = VP$. Suppose the physical parties have a partition into three correctable regions. Let T be a transversal unitary and $L \in U(q)$, with $E(T, L) \leq \varepsilon$. For all logical Weyl operators P, Q,

$$q^{-1/2} \min_{|z|=1} \|[[L, P], Q] - zI\|_{\mathrm{HS}} \leq 8\varepsilon, \tag{6.1}$$

where $[A, B] = ABA^\dagger B^\dagger$.

Proof. Let R_1, R_2, R_3 be the assumed correctable partition. The stabilizer cleaning lemma supplies a representative $\widetilde{P}$ of P outside R_1 and a representative $\widetilde{Q}$ of Q outside R_2. For reference, cleaning here is the elementary symplectic statement that the restriction of a normalizer label to a correctable region lies in the restriction of the stabilizer label space; subtracting that stabilizer removes the restriction.

Transversality makes $[T, \widetilde{P}]$ supported outside R_1, so $C = [[T, \widetilde{P}], \widetilde{Q}]$ is supported on $R_1^c \cap R_2^c = R_3$. The composition rules give

$$E(C, [[L, P], Q]) \leq 4\varepsilon. \tag{6.2}$$

Knill–Laflamme correction on R_3 gives $V^\dagger C V = cI$. Hence the logical nested commutator lies within 4ε of cI. Its normalized Hilbert–Schmidt norm is one, so $|c| \geq 1 - 4\varepsilon$. If $c = 0$, then $\varepsilon \geq 1/4$, and (6.1) is immediate because its left-hand side is at most $2 \leq 8\varepsilon$. Otherwise, replacing c by $c/|c|$ proves (6.1). Notice that C need not preserve the code. Only its scalar compression is used, while (6.2) controls leakage. $\square$

The remaining step is finite-dimensional Fourier concentration. It is stated separately because it is the point where scalar commutators become Clifford axes.

Lemma 6.2 (Nested Weyl commutators round to a Clifford)**.** Let $L \in U(q)$, $q = p^e$, and suppose for all Weyl labels v, w that

$$q^{-1/2} \min_{|z|=1} \|[[L, W_v], W_w] - zI\|_{\mathrm{HS}} \leq \eta.$$

If

$$\eta < \frac{1}{3},$$

then some additive one-qudit Clifford K satisfies

$$q^{-1/2} \min_{|z|=1} \|L - zK\|_{\mathrm{HS}} \leq \eta.$$

Proof. Put $A_v = L W_v L^\dagger$. Conjugation of W_w by W_v changes only its phase, so every $[A_v, W_w]$ is within η of a scalar. We use the following Fourier observation twice. If $A = \sum_a c_a W_a$ is unitary and every $[A, W_w]$ is within η of a scalar, then A lies within η of one Weyl axis. Indeed, with $p_a = |c_a|^2$, the commutator estimate gives

$$\left| \sum_a p_a \chi_w(a) \right| \geq 1 - \eta^2/2.$$

Character orthogonality, averaged over w, yields $\sum_a p_a^2 \geq (1 - \eta^2/2)^2$. Thus $\max_a p_a \geq (1 - \eta^2/2)^2$, which is precisely the asserted normalized distance bound.

Apply the observation to each A_v, and call its unique nearby Weyl label $F(v)$. The exact identity $A_{v+u} \in \mathbb{C} A_v A_u$ puts the axes labelled by $F(v+u)$ and $F(v)+F(u)$ within 3η. Distinct Weyl axes have phase-optimized distance $\sqrt{2}$, so $\eta < 1/3$ makes F additive. Comparing the exact commutator phases is equally direct. Write $\|X\|_2 = q^{-1/2}\|X\|_{\mathrm{HS}}$, and choose phases α, β so that $\|A_v - \alpha W_{F(v)}\|_2 \leq \eta$ and $\|A_u - \beta W_{F(u)}\|_2 \leq \eta$. Unitary invariance and a four-factor telescoping estimate give

$$\begin{aligned} &\big\|A_v A_u A_v^\dagger A_u^\dagger - W_{F(v)} W_{F(u)} W_{F(v)}^\dagger W_{F(u)}^\dagger\big\|_2 \\ &\qquad \leq \|A_v - \alpha W_{F(v)}\|_2 + \|A_u - \beta W_{F(u)}\|_2 + \|A_v^\dagger - \overline{\alpha} W_{F(v)}^\dagger\|_2 + \|A_u^\dagger - \overline{\beta} W_{F(u)}^\dagger\|_2 \leq 4\eta. \end{aligned}$$

The phases α, β cancel in the commutator. Its two sides are scalar p-th roots of unity: the first has the original Weyl commutator phase, while the second has the phase determined by

$F(v), F(u)$. Put $d = [F(v), F(u)] - [v, u] \in \mathbb{F}_p$ and $\omega = e^{2\pi i/p}$. Additivity makes F $\mathbb{F}_p$-linear, so applying the same estimate to (tv, u) gives $|1 - \omega^{td}| \le 4\eta$ for every $t \in \mathbb{F}_p$. If $d \neq 0$, character orthogonality gives

$$2 = \frac{1}{p} \sum_{t \in \mathbb{F}_p} |1 - \omega^{td}|^2 \le 16\eta^2 < \frac{16}{9} < 2,$$

a contradiction. Thus F preserves the trace-symplectic form. Nondegeneracy then makes F injective, hence invertible.

Choose an additive Clifford K_0 inducing F. Then $K_0^\dagger L$ approximately fixes every Weyl axis, so all its Weyl commutators are within η of scalars. The Fourier observation puts it within η of a Weyl; multiplying that Weyl into K_0 gives the required Clifford. $\square$

Two further estimates turn local Clifford frames into the stated global decomposition. The first separates distinct Clifford images of a stabilizer state by a uniform gap.

Lemma 6.3 (Quantized stabilizer overlap)**.** Let $|\psi\rangle, |\phi\rangle$ be stabilizer states on n qudits of local dimension $q = p^e$, with label groups L_ψ, L_ϕ and stabilizer characters χ_ψ, χ_ϕ. Then

$$|\langle \psi | \phi \rangle|^2 = \begin{cases} |L_\psi \cap L_\phi| / q^n, & \text{if } \chi_\psi = \chi_\phi \text{ on } L_\psi \cap L_\phi, \\ 0, & \text{otherwise.} \end{cases}$$

In particular, a product Clifford K either preserves the ray of ψ, or

$$\varepsilon(K) \ge d_p := \left(2 - 2p^{-1/2}\right)^{1/2}.$$

Proof. Weyl orthogonality and the stabilizer expansions give

$$|\langle \psi | \phi \rangle|^2 = q^{-n} \sum_{\ell \in L_\psi \cap L_\phi} (\chi_\psi \overline{\chi_\phi})(\ell).$$

The summand is a character of the intersection, so the sum is its order when the character is trivial and zero otherwise. Since the intersection has p-power order, every nonunit overlap has modulus at most $p^{-1/2}$. Apply this to $\phi = K\psi$, which is again a stabilizer state. $\square$

The second estimate controls all residual generators at once. Its proof is included because it is a load-bearing part of the headline quantitative theorem. Appendix B instead explains the infinitesimal stability supplied by pairwise maximal mixing alone.

Lemma 6.4 (Balanced-cut comparison)**.** Let $|\psi\rangle$ be a stabilizer AME$(2m, q)$ state with label group L, let $U = \bigotimes_j U_j$ be any product unitary, and write $a_j(v) = q^{-1} \operatorname{Tr}(W_v^\dagger U_j)$. For an m-party set A, define vectors indexed by L by

$$u_A(\ell) = \prod_{j \in A} |a_j(\ell_j)|, \qquad v_A(\ell) = \prod_{j \notin A} |a_j(\ell_j)|.$$

Then $\|u_A\| = \|v_A\| = 1$ and

$$|\langle \psi | U | \psi \rangle| \le \langle u_A, v_A \rangle, \qquad \varepsilon(U)^2 \ge \|u_A - v_A\|^2.$$

Proof. Restriction of L to either half is injective because a kernel label would be supported on at most m parties; it is bijective because both groups have order q^{2m}. Weyl Parseval then gives the unit norms. Expanding the stabilizer expectation and taking absolute values termwise gives $|\langle \psi | U | \psi \rangle| \le \langle u_A, v_A \rangle$; the Weyl index is $-\ell$, up to phase, and $\ell \mapsto -\ell$ is a bijection of L. Since the two vectors are unit, the asserted defect bound follows. $\square$

Proposition 6.5 (Residual stability)**.** Let $|\psi\rangle$ be a stabilizer AME$(2m,q)$ state with $m \geq 2$, and let $U = \bigotimes_j U_j$ be any product unitary. Put

$$\eta_j = 1 - \left|q^{-1}\operatorname{Tr} U_j\right|^2, \qquad H = \sum_j \eta_j.$$

If $H \leq 1/4$, then

$$\varepsilon(U)^2 \geq H/4.$$

Consequently, suppose $U = \bigotimes_j e^{ih_j}$, where every h_j is traceless Hermitian of spectral spread at most $s \leq \pi$, and put

$$c(s) = \max\left\{1 - \frac{s^2}{12}, \frac{4}{\pi^2}\right\}.$$

If $D^2 = \sum_j \|h_j\|_F^2 \leq q/4$, then

$$D \leq 2\sqrt{q/c(s)}\,\varepsilon(U) \leq \pi\sqrt{q}\,\varepsilon(U).$$

No bound on $\sum_j \|h_j\|_{\mathrm{op}}$ is assumed.

Proof. Weyl amplitudes and the balanced cut. Choose an m-party set A, use the notation of Lemma 6.4, and abbreviate $u = u_A$, $v = v_A$. Then $\eta_j = 1 - |a_j(0)|^2$, and the lemma gives $\varepsilon(U)^2 \geq \|u - v\|^2$.

Minimum-support witnesses. For each $j \notin A$, labels supported on $A \cup \{j\}$ form the kernel of projection to $A^c \setminus \{j\}$. Projection onto A^c is a bijection, so this kernel has order q^2. Moreover, evaluation at any coordinate in $A \cup \{j\}$ is injective: an element in its kernel would be supported on at most m parties and hence would vanish by the AME support formula. Evaluation is therefore bijective, and every nonzero label in the kernel has support exactly $A \cup \{j\}$.

For a Weyl label w, let $\ell^{j,w}$ be the unique such label whose j-coordinate is w. Put $\mu_i(v) = |a_i(v)|^2$. The first sum below is exact, while the second is the diagonal part of a product of nonnegative sums:

$$\sum_{w\neq 0} v(\ell^{j,w})^2 = \eta_j \prod_{k\in A^c\setminus\{j\}} (1-\eta_k) \geq \eta_j(1-H),$$
$$\sum_{w\neq 0} u(\ell^{j,w})^2 = \sum_{w\neq 0}\prod_{i\in A} \mu_i(\ell_i^{j,w}) \leq \prod_{i\in A}\sum_{w\neq 0} \mu_i(\ell_i^{j,w}) = \prod_{i\in A} \eta_i.$$

The corresponding families are disjoint as j varies. The families obtained after exchanging A and A^c also have different supports, so their contributions may be added. Applying $(x-y)^2 \geq x^2/2 - y^2$, first to the families indexed by $j \in A^c$ and then to those indexed by $i \in A$, gives

$$\|u-v\|^2 \geq \tfrac{1}{2}(1-H)H - m\left(\prod_{i\in A}\eta_i + \prod_{j\in A^c}\eta_j\right).$$

For either half S, $m\prod_{i\in S}\eta_i \leq \eta^{m-1}\sum_{i\in S}\eta_i$, where $\eta = \max_j \eta_j$. Hence

$$\|u-v\|^2 \geq \left(\tfrac{1}{2}(1-H) - \eta^{m-1}\right)H.$$

For $m \geq 3$ and $H \leq 1/4$, the coefficient is at least $5/16$. For $m = 2$, write $A = \{1,2\}$, $A^c = \{3,4\}$, and $P = \eta_1\eta_2 + \eta_3\eta_4$. Using the exact first identity above in the two cut directions gives

$$\|u-v\|^2 \geq H/2 - 3P,$$

while

$$P \leq \frac{(\eta_1+\eta_2)^2+(\eta_3+\eta_4)^2}{4} \leq \frac{H^2}{4}.$$

Therefore $H/2 - 3P \geq H/2 - 3H^2/4 \geq 5H/16$. Thus in all cases $\varepsilon(U)^2 \geq 5H/16$. We use the weaker $H/4$ bound to simplify the subsequent constants.

Spectral contraction. It remains to compare H with D. If X is uniform on the eigenvalues of h_j, then

$$\eta_j = 1 - |\mathbb{E}e^{iX}|^2 = \mathbb{E}[1-\cos(X-X')] \geq c(s)\frac{\|h_j\|_F^2}{q},$$

using both $\cos x \leq 1 - x^2/2 + x^4/24$ for $|x| \leq s$ and $1-\cos x \geq 2x^2/\pi^2$ for $|x| \leq \pi$. Conversely $\eta_j \leq \|h_j\|_F^2/q$, so $H \leq D^2/q \leq 1/4$. Combining the inequalities gives $\varepsilon(U)^2 \geq c(s)D^2/(4q)$, which proves the consequence. □

At $\varepsilon = 0$, the first two lemmas in this section recover the three-cleanable-region Clifford conclusion of [PY15, Lemma 5]. Their quantitative combination, and the constants in (6.1), are what will be used here.

The next elementary estimate records the conversion from a nearby unitary axis to a centered Hermitian generator.

Lemma 6.6 (Chords and centered logarithms)**.** Let $V \in U(q)$, and put

$$\delta = q^{-1/2}\min_{|z|=1}\|V - zI\|_{\mathrm{HS}}.$$

If $\delta < \sqrt{2/q}$, choose a minimizing phase z, write $z^{-1}V = e^{ia}$ using the principal Hermitian logarithm, and set $h = a - q^{-1}\operatorname{Tr}(a)I$. Then the spectrum of a lies in $(-\pi/2, \pi/2)$, the spectral spread of h is at most π, and

$$\|h\|_F \leq \|a\|_F \leq \frac{\pi}{2}\sqrt{q}\,\delta.$$

Proof. Since the operator norm is at most the Hilbert–Schmidt norm,

$$\|z^{-1}V - I\|_{\mathrm{op}} \leq \|z^{-1}V - I\|_{\mathrm{HS}} = \sqrt{q}\,\delta < \sqrt{2}.$$

Thus every eigenvalue $e^{i\theta}$ satisfies $|e^{i\theta}-1| < \sqrt{2}$, so its principal angle obeys $|\theta| < \pi/2$. Centering is orthogonal projection onto the traceless Hermitian matrices and does not change spectral spread. Finally, $|\theta| \leq (\pi/2)|e^{i\theta}-1|$ for $|\theta| \leq \pi/2$; summing its square over the eigenvalues proves the norm bound. □

Theorem 6.7 (Cleaning-based global rounding)**.** Let $|\psi\rangle$ be a stabilizer AME$(2m, q)$ state, $q = p^e$, $m \geq 2$, and $n = 2m$. Let $U = \bigotimes_i U_i$ be a product unitary, and put

$$R_{n,q}^{\mathrm{clean}} = \min\left\{\frac{1}{4\sqrt{2q}}, \frac{1}{8\pi\sqrt{n}}\right\}. \tag{6.3}$$

If $\varepsilon(U) < R_{n,q}^{\mathrm{clean}}$, then, up to a global phase,

$$U = g\bigotimes_{i=1}^{n} e^{ih_i}, \qquad g \in G(\psi),$$

where the h_i are traceless Hermitian, have spectral spread at most π, and satisfy

$$\left(\sum_i \|h_i\|_F^2\right)^{1/2} \leq \pi\sqrt{q}\,\varepsilon(U). \tag{6.4}$$

Before the exact-symmetry step, the recovered local Cliffords obey

$$q^{-1/2} \min_{|z|=1} \|U_i - zK_i\|_{\mathrm{HS}} \le 8\varepsilon(U) \tag{6.5}$$

at every party. For fixed q, (6.3) has order $n^{-1/2}$, conditional on existence of the stated AME states.

Proof. Local Clifford frames. Choose party i as the logical leg and write the AME Choi form as $q^{-1/2}\sum_a |a\rangle \otimes V_i|a\rangle$. The other local factors form a transversal physical unitary T_i. Choose z attaining $\varepsilon(U) = \|U\psi - z\psi\|$. Vectorization gives

$$L_i = z(U_i^{\mathsf{T}})^{-1}, \qquad E(T_i, L_i) = \varepsilon(U).$$

These encoders are Pauli-compatible by the Choi argument in Section 4. Since $n \ge 4$, (6.3) implies $8\varepsilon(U) < 1/(\pi\sqrt{n}) < 1/3$. Lemmas 6.1 and 6.2, with $\eta = 8\varepsilon(U)$, give (6.5); transpose and inverse preserve the additive Clifford group.

Centered residual generators. Put $K = \bigotimes_i K_i$. For each i, choose a phase z_i attaining the minimum in (6.5), and let a_i be the principal Hermitian logarithm defined by

$$z_i^{-1} K_i^\dagger U_i = e^{ia_i}.$$

Set $\bar{a}_i = q^{-1}\operatorname{Tr}(a_i)$ and $h_i = a_i - \bar{a}_i I$. Rephase K_i by $z_i e^{i\bar{a}_i}$; this changes $K = \bigotimes_i K_i$ only by a global phase and does not change its projective Clifford action. After this rephasing, $K_i^\dagger U_i = e^{ih_i}$, each h_i is traceless, centering does not increase its Frobenius norm, and its spectral spread is unchanged. By the first clause of (6.3), the bound (6.5) satisfies the hypothesis of Lemma 6.6; hence every spectral spread is at most π. Let $D^2 = \sum_i \|h_i\|_F^2$. The same lemma gives

$$D^2 \le \frac{\pi^2 q}{4} \sum_i \left(q^{-1/2} \min_{|z|=1} \|U_i - zK_i\|_{\mathrm{HS}} \right)^2 \le 16\pi^2 qn\varepsilon(U)^2. \tag{6.6}$$

The second clause of (6.3) gives $D \le \sqrt{q}/2$.

Exact branch selection. Put $M = \sum_i h_i^{(i)}$, so that $K^\dagger U = e^{iM}$. Two-uniformity and tracelessness give

$$\|M\psi\|^2 = \langle\psi|M^2|\psi\rangle = D^2/q.$$

The integral identity $(e^{iM} - I)\psi = i\int_0^1 e^{itM} M\psi\, dt$ therefore gives

$$\|(K^\dagger U - I)\psi\| \le D/\sqrt{q} \le 4\pi\sqrt{n}\,\varepsilon(U).$$

The triangle inequality and $n \ge 4$ give

$$\varepsilon(K) \le (1 + 4\pi\sqrt{n})\varepsilon(U) < \frac{1}{2} + \frac{1}{8\pi\sqrt{n}} < \frac{3}{4} < d_2 \le d_p.$$

Thus Lemma 6.3 makes K an exact symmetry. Apply Proposition 6.5 to obtain (6.4). □

The theorem already contains a genuinely collective estimate once the exact branch has been selected. Writing $g = \bigotimes_i g_i$ in the factorization above,

$$\sum_i \left(q^{-1/2} \min_{|z|=1} \|U_i - zg_i\|_{\mathrm{HS}} \right)^2 \le \frac{D^2}{q} \le \pi^2 \varepsilon(U)^2. \tag{6.7}$$

The first inequality is the elementary chord bound $\|e^{ih_i} - I\|_{\mathrm{HS}} \le \|h_i\|_F$. Thus the $\sqrt{n}$ loss is absent after exact-symmetry recovery.

The same estimate applies to intertwiners between two states as soon as one exact intertwiner is fixed. This separates the cost of finding the correct orbit from stability within that orbit.

Corollary 6.8 (Relative intertwiner rounding)**.** Let $|\psi\rangle, |\phi\rangle$ be stabilizer AME$(2m, q)$ states as in Theorem 6.7, and suppose that $B = \bigotimes_i B_i$ is an exact product-unitary intertwiner from ψ to ϕ. For a product unitary $U = \bigotimes_i U_i$, put

$$\varepsilon_{\psi\to\phi}(U) = \min_{|z|=1} \|U\psi - z\phi\|.$$

If $\varepsilon_{\psi\to\phi}(U) < 1/24$, then each U_i is within normalized Hilbert–Schmidt distance $8\varepsilon_{\psi\to\phi}(U)$ of B_iK_i for a one-qudit Clifford K_i. If, in addition, $\varepsilon_{\psi\to\phi}(U) < R^{\mathrm{clean}}_{n,q}$, then, up to phase,

$$U = H \bigotimes_i e^{ih_i}, \qquad H\psi \in \mathbb{C}\phi, \qquad \left(\sum_i \|h_i\|_F^2\right)^{1/2} \le \pi\sqrt{q}\,\varepsilon_{\psi\to\phi}(U),$$

where H is an exact product-unitary intertwiner and the h_i are traceless Hermitian of spectral spread at most π. After branch selection,

$$\sum_i \left(q^{-1/2} \min_{|z|=1} \|U_i - zH_i\|_{\mathrm{HS}}\right)^2 \le \pi^2 \varepsilon_{\psi\to\phi}(U)^2.$$

Thus the stability constants are independent of the party count; the certified radius remains $R^{\mathrm{clean}}_{n,q}$.

Proof. Set $W = B^\dagger U = \bigotimes_i B_i^\dagger U_i$. Since $B\psi$ and ϕ span the same ray, $\varepsilon(W) = \varepsilon_{\psi\to\phi}(U)$. The first paragraph of the proof of Theorem 6.7 gives the local 8ε estimate under $\varepsilon(W) < 1/24$. Under the stronger radius hypothesis, apply the theorem and (6.7) to W. If $W = g \bigotimes_i e^{ih_i}$ with $g \in G(\psi)$, then $H = Bg$ is an exact intertwiner from ψ to ϕ, and left multiplication by each B_i preserves normalized Hilbert–Schmidt distance. This proves all three estimates. Exact rigidity also makes every B_i Clifford, but that fact is not needed for the reduction. $\square$

This does not enlarge the entry radius in (6.3). Before the branch is known, cleaning supplies only the separate bounds (6.5) for locally chosen Cliffords K_i. The n encoded implementation errors are unitary reshapings of the same global defect vector, not projections onto orthogonal sectors, so their squared bounds cannot be summed by a Bessel argument. A radius-improving collective estimate would itself imply constant-radius branch selection: the chord–angle inequality would make $\sum_i \|h_i\|_F^2 = O(q\varepsilon^2)$, and the uniform stabilizer gap would force $\bigotimes_i K_i$ to be exact. The missing input is therefore compatibility of the recovered discrete Clifford data, not a sharper final norm summation.

The linear part of that compatibility can in fact be recovered without a party-count or local-dimension loss.

Proposition 6.9 (Robust compatibility of the transition maps)**.** Let $|\psi\rangle$ be a stabilizer AME$(2m, q)$ state, $q = p^e$, $m \ge 2$, and let $U = \bigotimes_i U_i$ be a product unitary. Suppose

$$\varepsilon(U) < R^{\mathrm{trans}} := \frac{\sqrt{2}}{68}.$$

The cleaning and Fourier lemmas give local Cliffords K_i satisfying (6.5). Let F_i be their symplectic maps. Then

$$F_j M_A(i,j) = M_A(i,j) F_i$$

for every $(m+1)$-set A and $i, j \in A$. Consequently $F = \bigoplus_i F_i$ preserves the stabilizer label group L, and some product-Pauli correction W_vK is an exact symmetry of ψ. For any prescribed m-party set, there is a unique such correction label v supported on that set.

Proof. Fix A, distinct $i, j \in A$, and put $B = A^c \cup \{i, j\}$. Both A and B have size $m+1$, while $A \cap B = \{i, j\}$. Choose $\ell \in L(A)$ and $r \in L(B)$. Using the stabilizer lifts $\widetilde{W}_\ell$ and $\widetilde{W}_r$ from (2.2), set

$$C = [U\widetilde{W}_\ell U^\dagger, \widetilde{W}_r] = [UW_\ell U^\dagger, W_r],$$

where the second equality holds because scalar lift phases cancel from a commutator. After choosing the phase of U realizing its defect, $\|(U\widetilde{W}_\ell U^\dagger - I)\psi\| \leq 2\varepsilon(U)$, and the same holds with the adjoint. Since $\widetilde{W}_r\psi = \psi$,

$$\|(C - I)\psi\| \leq 4\varepsilon(U). \tag{6.8}$$

The commutator is supported only on $A \cap B = \{i, j\}$. Since $R^{\text{trans}} < 1/24$, the cleaning and Fourier lemmas apply. Choose phases so that $\delta_s = q^{-1/2}\|U_s - K_s\|_{\text{HS}} \leq 8\varepsilon(U)$. Replacing $U_s W_{\ell_s} U_s^\dagger$ by its K_s-conjugate costs at most $2\delta_s$ in normalized Hilbert–Schmidt norm, and replacing its commutator with W_{r_s} costs at most $4\delta_s$. Let $C_K = [KW_\ell K^\dagger, W_r]$. Both commutators are supported on $S = \{i, j\}$; telescoping their two unitary factors and using $\rho_S = q^{-2}I$ gives

$$\|(C - C_K)\psi\| = q^{-1}\|C - C_K\|_{\text{HS},S} \leq 4(\delta_i + \delta_j) \leq 64\varepsilon(U).$$

Here the Hilbert–Schmidt norm is taken on the two-site space. The Clifford commutator is a scalar p-th root ζ, so (6.8) gives $|1 - \zeta| \leq 68\varepsilon(U)$. Replacing ℓ by $t\ell$, for every $t \in \mathbb{F}_p$, leaves these bounds unchanged and replaces ζ by ζ^t, because the Clifford label maps and the commutator pairing are $\mathbb{F}_p$-linear. If $\zeta \neq 1$, character orthogonality would give

$$2 = \frac{1}{p}\sum_{t \in \mathbb{F}_p} |1 - \zeta^t|^2 \leq (68\varepsilon(U))^2 < 2.$$

Hence $\zeta = 1$.

Write $a = \ell_i$, $b = r_i$, $\ell_j = M_A(i, j)a$, and $r_j = M_B(i, j)b$. The identity $\zeta = 1$, for all a, b, says

$$[F_i a, b] + [F_j M_A(i, j)a, M_B(i, j)b] = 0. \tag{6.9}$$

Here brackets denote the trace-symplectic pairing. Isotropy of L, applied to the element of $L(A)$ whose i-coordinate is $F_i a$, gives

$$[F_i a, b] + [M_A(i, j)F_i a, M_B(i, j)b] = 0.$$

Subtracting this from (6.9), using surjectivity of $M_B(i, j)$ and nondegeneracy of the pairing, proves the transition equation. The exact transition-map theorem gives $FL = L$, and Lemma 4.2 supplies W_v. $\square$

The uncontrolled Pauli correction does not obstruct the logical action of a chosen encoder.

Corollary 6.10 (Logical Clifford rounding)**.** Under the hypotheses and radius of Proposition 6.9, choose party i as the logical leg, and let L_i be the Choi-associated logical unitary approximately implemented by the other $2m-1$ factors as in the proof of Theorem 6.7. There is an exactly transversally realizable logical Clifford L_i' such that

$$q^{-1/2}\min_{|z|=1}\|L_i - zL_i'\|_{\text{HS}} \leq 8\varepsilon(U).$$

Proof. Write the exact symmetry supplied by the proposition as $W_v K$, where $K = \bigotimes_j K_j$. The projection of the stabilizer label group onto the Weyl plane at party i is surjective: this already holds for every minimum-support subgroup containing i. Multiply $W_v K$ by a stabilizer whose i-th label is $-v_i$. The result is still an exact product-Clifford symmetry, and its factor at i is projectively K_i. The Choi correspondence therefore makes its other factors an exact transversal implementation of $L_i' = (K_i^\top)^{-1}$. The logical unitary L_i is the same transpose–inverse transform of U_i, up to phase, so (6.5) proves the displayed bound. □

The Pauli correction in the proposition is essential. All localized commutators are unchanged when a recovered K_i is multiplied by a local Weyl: that multiplication changes conjugation phases, and the commutator cancels them. Equivalently, the transition maps see L but not its stabilizer character. The correction v can therefore be nonzero modulo L, in which case $W_v K$ need not be locally close to U. Its unique half-supported normal form localizes the ambiguity but does not bound its distance from the identity. Exact affine compatibility cannot be obtained from these commutator tests, and Proposition 6.9 alone does not improve (6.3). Controlling the character discrepancy is the same remaining branch-selection problem isolated by the collective estimate above.

Uniformly over prime powers $q = p^e$ and existing tensors,

$$R_{n,q}^{\mathrm{clean}} = \Theta\Big(\min\{q^{-1/2}, n^{-1/2}\}\Big)\,, \tag{6.10}$$

with absolute implied constants. Character averaging removes any separate dependence on the characteristic. An asymptotic use of (6.10) must still specify an existing family.

For the generalized and extended generalized Reed–Solomon construction [SR86], together with the standard MDS-to-AME construction [RGRA18], $n = 2m \le q + 1$. Hence

$$R_{n,q}^{\mathrm{clean}} = \Theta(q^{-1/2}) \tag{6.11}$$

over both prime and extension fields. On a length-saturating subfamily $n \asymp q$, this is $\Theta(n^{-1/2})$. The fixed-q, $n^{-1/2}$ reading of the minimum is conditional on existence; it is not an AME-existence claim.

The cleaning radius is the governing bound. Its $n^{-1/2}$ scale is a proved lower bound, not an optimality claim; no product-unitary family is known to saturate the collective frame-error estimate (6.6).

6.1 An observable sufficient condition

The defect in the rounding theorem can be bounded from the marginal tests of Proposition 4.6. No reconstruction of the local unitaries is needed to establish this entry condition.

Corollary 6.11 (Marginal-certified rounding)**.** Let ψ be a stabilizer AME$(2m, q)$, $m \ge 2$, and suppose the tested state is $U|\psi\rangle\langle\psi|U^\dagger$ for a product unitary U. Put $R = R_{2m,q}^{\mathrm{clean}}$ and $\nu_* = (m+1)/(2m)$. If its rejection probability satisfies

$$r \le \gamma < \nu_*(R^2 - R^4/4),$$

then Theorem 6.7 applies with defect bound

$$\varepsilon(U) \le \eta(\gamma) := \sqrt{2\big(1 - \sqrt{1 - \gamma/\nu_*}\big)} < R.$$

In particular, every factor rounds within $8\eta(\gamma)$, and the collective residual bound is $\pi\sqrt{q}\,\eta(\gamma)$.

Proof. Write $F = |\langle\psi|U|\psi\rangle|^2$. Equation (4.2) gives $F \ge 1 - \gamma/\nu_*$, and $\varepsilon(U)^2 = 2(1 - \sqrt{F})$. Since $0 < R < \sqrt{2}$, the assumed bound on γ gives $\eta(\gamma) < R$. □

Here γ must be a valid upper bound on the rejection probability, for example a statistical confidence bound under the chosen sampling assumptions; a bare observed failure frequency is not such a bound. For arbitrary mixed preparations the tests still certify fidelity, but the Clifford conclusion requires the product-unitary promise.

7 Conclusion

The same support geometry controls the exact and approximate problems. On an $(m+1)$-party marginal it exposes every local Weyl axis, which turns arbitrary product-unitary intertwiners into product Cliffords. The transition maps between the resulting local Weyl frames organize the residual freedom. After one party is reinterpreted as a logical input, the support geometry becomes three-region correctability. Cleaning and Weyl–Fourier concentration then recover the same discrete Clifford frame without paying the exponentially small amplitude of a single marginal.

The exact classification is finite and constructive. A half-set direct sum reduces the data to the dimensionally optimal m minimum supports and gives fixed-label recognition in polynomial time for fixed local dimension. Over prime fields, the common centralizer of the fundamental cycles is the intrinsic algebra of block-diagonal one-party label endomorphisms preserving the stabilizer. Its five possible types determine both the symmetry group and the additional code structure. For four and six parties this algebra is necessarily nonscalar. The same recognition procedure decides transversal conversion between the associated encoders and stochastic conversion between the states, since every product operator between two such states is Clifford up to scalars. The same m minimum supports carry the reductions that determine the state among all density operators, at the smallest locality any parent Hamiltonian can have. Testing both complementary families raises the verification gap to $(m+1)/(2m)$. This turns the marginal constraints into a fidelity certificate and, under a product-unitary preparation promise, an observable entry condition for robust rigidity.

The quantitative conclusion is uniform only when all varying parameters are displayed:

$$R_{n,q}^{\text{clean}} = \Theta\Big(\min\{q^{-1/2}, n^{-1/2}\}\Big), \qquad q = p^e.$$

For fixed local dimension this gives the conditional $n^{-1/2}$ scale along any existing stabilizer-AME family. On generalized and extended Reed–Solomon constructions, where $n \leq q+1$, the scale is $\Theta(q^{-1/2})$ over both prime and extension fields. These are certified radii; their optimal order is open.

The robust transition-map result identifies the remaining bottleneck. Linear symplectic compatibility becomes exact for $\varepsilon < \sqrt{2}/68$, independently of party count and local dimension, but commutators discard the affine Weyl phases. They therefore determine the stabilizer label group and not its character. The logical action of any chosen encoder can nevertheless be rounded within 8ε to an exactly transversally realizable Clifford, because a stabilizer cancels the Pauli correction on the logical leg. A party-count-independent global entry theorem would need a new way to control that character discrepancy, or an equivalent collective branch selection principle, before the product-Pauli correction is known to be locally small.

A Module structures from the endomorphism algebra

This appendix proves the last column of Table 1. It is not needed for LU rigidity, recognition, or robust rounding. Throughout, $m \geq 2$, $q = p$, and $E = \mathcal{E}_L$.

For every half-set S, projection $L \to \bigoplus_{i \in S} P_i$ is an E-linear isomorphism. Hence the E-module L is the direct sum of m copies of the local module.

If $E \cong \mathbb{F}_q \times \mathbb{F}_q$, its primitive idempotents split every local plane into two lines and split $L = L_X \oplus L_Z$. Half-set bijectivity restricts to each summand, so L_X and L_Z are classical $[2m, m, m+1]_q$ MDS codes. After compatible local symplectic bases are chosen, the ambient alternating form pairs the two axis systems; isotropy and dimension identify the codes as weighted duals. This gives an MDS–CSS stabilizer after local symplectic coordinate changes.

If $E \cong \mathbb{F}_{q^2}$, every local plane is its one-dimensional regular module. Thus $L \cong E^m$, and the half-set isomorphisms make it an E-linear $[2m, m, m+1]_{q^2}$ MDS code. In compatible local E-coordinates, every nondegenerate alternating $\mathbb{F}_q$-form on E has the form

$$(x, y) \longmapsto \mathrm{Tr}_{E/\mathbb{F}_q}(\kappa_i x y^q), \qquad \kappa_i^q = -\kappa_i \neq 0. \tag{A.1}$$

Applying isotropy to ax, y for all $a \in E$, then using the nondegenerate trace pairing, gives

$$\sum_i \kappa_i x_i y_i^q = 0 \qquad (x, y \in L).$$

The trace-zero elements form a one-dimensional $\mathbb{F}_q$-space. After one nonzero trace-zero κ is fixed, write $\kappa_i = w_i \kappa$ with $w_i \in \mathbb{F}_q^\times$. The displayed relation then becomes $\sum_i w_i x_i y_i^q = 0$ after division by κ, which is weighted Hermitian self-duality. The argument includes characteristic two.

If $E \cong \mathbb{F}_q[\epsilon]/(\epsilon^2)$, a faithful two-dimensional $\mathbb{F}_q$-module has a nonzero square-zero action and is therefore the regular rank-one module. Hence L is free of rank m over E, and every half-set projection is an isomorphism of free modules. No distance or self-duality convention over the ring is used.

Finally, if $E = M_2(\mathbb{F}_q)$, align the local actions by local module coordinates. The unique simple module is $\mathbb{F}_q^2$, and semisimplicity gives

$$L = \mathbb{F}_q^2 \otimes C$$

for a multiplicity space $C \subseteq \mathbb{F}_q^{2m}$. Dimension and half-set bijectivity make C a classical $[2m, m, m+1]_q$ MDS code. Each local alternating form is $w_i J$ in these coordinates, with $w_i \neq 0$, and isotropy becomes

$$\sum_i w_i c_i d_i = 0 \qquad (c, d \in C).$$

Thus C is self-dual for the weighted coordinate pairing, giving the maximally symmetric MDS–CSS row after local symplectic coordinate changes.

B Two-uniform discreteness and local stability

This appendix isolates what pairwise maximal mixing proves without any stabilizer hypothesis. It forces the projective product-unitary symmetry group to be finite and fixes the infinitesimal quadratic stability constant. These results explain the local geometry behind robust rigidity, but they do not identify the finite group or supply the explicit entry radius of Section 6. Readers interested only in the headline exact and quantitative theorems may skip this appendix.

B.1 Discreteness from two-party maximal mixing

The rigidity theorem uses the stabilizer hypothesis to identify the symmetry group. Finiteness of that group needs far less: two-party marginals alone, at any local dimension, with no stabilizer structure. Throughout this subsection $q \geq 2$ is an arbitrary integer; only the statements naming Clifford operators return to $q = p^e$.

Call a pure state $|\psi\rangle \in (\mathbb{C}^q)^{\otimes n}$ *2-uniform* if

$$\rho^{\psi}_{\{j,k\}} = q^{-2} I \qquad \text{for all } j \neq k.$$

Tracing once more makes every one-party marginal $q^{-1}I$. Such states exist only for $n \geq 4$, since at $n = 3$ purity forces $\operatorname{rank}\rho_{\{1,2\}} = \operatorname{rank}\rho_{\{3\}} \leq q$; and every AME$(2m, q)$ state with $m \geq 2$ is 2-uniform, stabilizer or not. Write

$$G(\psi) = \{U_1 \otimes \cdots \otimes U_n : (U_1 \otimes \cdots \otimes U_n)|\psi\rangle \in \mathbb{C}|\psi\rangle\}$$

for its product-unitary symmetry group and, for a product unitary U,

$$\varepsilon(U) = \min_\theta \big\| U|\psi\rangle - e^{i\theta}|\psi\rangle \big\| = \big(2 - 2|\langle\psi|U|\psi\rangle|\big)^{1/2}$$

for its defect. Because operators on distinct sites commute, a product unitary is a single exponential: $\bigotimes_j e^{iH_j} = e^{iL}$ with $L = \sum_j H_j^{(j)}$, where $H^{(j)}$ denotes H acting at site j. Splitting each H_j into a traceless part h_j and a multiple of the identity leaves $L = M + cI$ with $M = \sum_j h_j^{(j)}$. Everything below rests on one computation about M.

Lemma B.1 (Local-generator isometry)**.** Let $|\psi\rangle$ be 2-uniform and let h_j and h'_j, $1 \leq j \leq n$, be traceless Hermitian operators on $\mathbb{C}^q$. Put $M = \sum_j h_j^{(j)}$ and $M' = \sum_j h_j'^{(j)}$. Then $\langle\psi|M|\psi\rangle = 0$ and

$$\big\langle M|\psi\rangle,\, M'|\psi\rangle\big\rangle = \frac{1}{q}\sum_{j=1}^{n} \operatorname{Tr}(h_j h'_j). \tag{B.1}$$

In particular $\|M|\psi\rangle\|^2 = q^{-1}\sum_j \|h_j\|_F^2$, so $(h_1, \ldots, h_n) \mapsto \sqrt{q}\, M|\psi\rangle$ is an isometry from the traceless local generators, carrying the product Frobenius form, into the Hilbert space regarded as a real inner product space.

Proof. One-party maximal mixing gives $\langle M\rangle = 0$. In the expansion of $\langle MM'\rangle$, equal-site terms are $\operatorname{Tr}(h_j h'_j)/q$, while distinct-site terms are $\operatorname{Tr}(h_j)\operatorname{Tr}(h'_k)/q^2 = 0$. Thus the displayed inner product has the stated real part. Its imaginary part is $\langle [M, M']\rangle/2i = 0$, because only traceless one-site commutators remain. $\square$

Pairwise maximal mixing is essential: for a Bell pair, $M = h \otimes I - I \otimes h^T$ annihilates the state, producing the familiar $U \otimes \overline{U}$ continuous symmetry.

We also need the standard Lie-algebra form of a product-unitary generator.

Lemma B.2 (Product generators)**.** Let $\pi : U(q)^n \to U(q^n)$ send $(U_1, \ldots, U_n)$ to $U_1 \otimes \cdots \otimes U_n$. Then π is a smooth homomorphism of compact Lie groups, its image P is closed, and

$$\operatorname{Lie}(P) = \Big\{ i\textstyle\sum_j H_j^{(j)} : H_j = H_j^\dagger \text{ on } \mathbb{C}^q \Big\}. \tag{B.2}$$

Thus every one-parameter subgroup of a closed subgroup of P has such a generator.

Proof. The source is compact, so P is closed. At the identity,

$$d\pi(iH_1, \ldots, iH_n) = i\sum_j H_j^{(j)}.$$

Surjectivity onto P makes this map onto $\operatorname{Lie}(P)$. The final claim is the closed-subgroup theorem. $\square$

Theorem B.3 (Discreteness from 2-uniformity)**.** Let $|\psi\rangle$ be a 2-uniform pure state of n qudits of local dimension q. Then $\operatorname{Lie}(G(\psi)) = i\mathbb{R}I$, and $G(\psi)/U(1) \cdot I$ is finite. In particular every AME$(2m, q)$ state with $m \geq 2$, stabilizer or not, has finite product-unitary symmetry group modulo global phase.

Proof. The set $\{U \in U(q^n) : U|\psi\rangle \in \mathbb{C}|\psi\rangle\}$ is a closed subgroup, so $G(\psi) = P \cap \{\cdots\}$ is a closed subgroup of P and hence a compact Lie group. Let $X \in \mathrm{Lie}(G(\psi))$. By Lemma B.2, $X = iL$ with $L = \sum_j H_j^{(j)}$ Hermitian; write $L = M + cI$ as above. Each e^{itL} fixes the ray of $|\psi\rangle$, so $e^{itL}|\psi\rangle = \lambda(t)|\psi\rangle$ with $\lambda(t) = \langle\psi|e^{itL}|\psi\rangle$ unimodular and smooth in t. Differentiating at $t = 0$ shows that $|\psi\rangle$ is an eigenvector of the Hermitian operator L, say $L|\psi\rangle = \mu|\psi\rangle$ with μ real. Pairing with $|\psi\rangle$ and using $\langle M\rangle = 0$ gives $\mu = c$, hence $M|\psi\rangle = 0$. By Lemma B.1, $\sum_j \|h_j\|_F^2 = 0$, so every $h_j = 0$ and every H_j is scalar. Thus $\mathrm{Lie}(G(\psi)) = i\mathbb{R}I$. Since $U(1)\cdot I$ is a closed central one-dimensional subgroup of $G(\psi)$, the compact Lie group $G(\psi)/U(1)\cdot I$ has dimension zero and is therefore finite. ☐

Remark B.4 (Which torus is divided out). Upstairs in $U(q)^n$ the preimage $\pi^{-1}(G(\psi))$ always contains the local-phase torus $T = U(1)^n$, of dimension n. This does not contradict the theorem: $\pi(T) = U(1)\cdot I$ and $\ker\pi \subset T$ has dimension $n-1$, so $\pi^{-1}(G(\psi))/T \cong G(\psi)/U(1)\cdot I$. Statements about local factors below are statements on $\mathrm{PU}(q)^n$, not statements modulo one global phase.

Remark B.5 (Discreteness in the literature). Wirthmüller proves related finiteness criteria for binary stabilizer codes [Wir11, Theorem 7 and Corollary 11]; qubit stabilizer symmetries are classified in [EK20], and Tan computes the four-qutrit AME group [Tan26b, Theorem 5.3]. The theorem above instead uses only 2-uniformity and allows arbitrary local dimension and nonstabilizer states.

The hypothesis cannot be reduced to one-party maximal mixing: such states can have positive-dimensional local-unitary symmetry [SSM21, Lemma 2]. Thus pairwise maximal mixing, not merely criticality of the state, is what eliminates continuous product symmetries.

Remark B.6 (Two-uniformity does not identify the group). The equal-phase CSS state of $C = \mathrm{RM}(1,4) \leq \mathbb{F}_2^{16}$ is 2-uniform because $d(C) = 8$ and $d(C^\perp) = 4$. Yet $T^{\otimes 16}$, $T = \mathrm{diag}(1, e^{i\pi/4})$, fixes it: all codeword weights are $0, 8$, or 16. Since T does not preserve Weyl axes, two-uniformity forces finiteness but not Cliffordness. The AME hypothesis and Theorem 1.1 supply that second conclusion.

Corollary B.7 (Discrete product-unitary symmetry)**.** For a stabilizer AME state as in Theorem 1.1, let $G \leq U(q)^{2m}$ be its local-factor tuple group and let $\widetilde{G} \leq U(q)^{2m} \rtimes S_{2m}$ also allow party permutations. The coordinatewise scalar torus $T = (S^1)^{2m}$ is the identity component of both groups, and $\Gamma = G/T$ and $\widetilde{\Gamma} = \widetilde{G}/T$ are finite discrete groups. If Π is the realized party-permutation group, then

$$1 \longrightarrow \Gamma \longrightarrow \widetilde{\Gamma} \longrightarrow \Pi \longrightarrow 1.$$

This extension splits exactly when the realized permutations admit a homomorphic choice of projective product-unitary lifts.

Proof. Theorem 1.1 puts every local factor in the finite projective Clifford group. Projectivization has kernel T, so its image is finite and discrete and T is the identity component. Retaining the party permutation defines the quotient map $\widetilde{\Gamma} \to \Pi$; its kernel is Γ, giving the displayed sequence. Its splitting criterion is the definition of a split extension. ☐

B.2 Local quadratic stability

Discreteness makes the exact symmetries isolated. The same identity $(B.1)$ says how fast the defect grows away from them, with a constant that does not involve the number of parties.

Theorem B.8 (Local stability)**.** Let $|\psi\rangle$ be 2-uniform, let each h_j be traceless Hermitian on $\mathbb{C}^q$, and put

$$U = \bigotimes_{j=1}^{n} e^{ih_j}, \qquad t = \sum_{j=1}^{n} \|h_j\|_{\mathrm{op}}, \qquad D = \Big(\sum_{j=1}^{n} \|h_j\|_F^2\Big)^{1/2}.$$

If $t \le \frac{1}{2}$ then

$$\frac{D^2}{q}\Big(1 - \frac{t}{3}\Big) \le \varepsilon(U)^2 \le \frac{D^2}{q}\Big(1 + \frac{t}{3}\Big). \tag{B.3}$$

In particular $D \le \sqrt{6q/5}\,\varepsilon(U) < 1.096\sqrt{q}\,\varepsilon(U)$, and $\varepsilon(U)^2/D^2 \to q^{-1}$ as $t \to 0$, uniformly in n.

Proof. Since the sites commute, $U = e^{iM}$ with $\|M\|_{\mathrm{op}} \le t$. Put $\mu_2 = \langle M^2\rangle = D^2/q$, using Lemma B.1. Taylor's theorem and the spectral calculus give

$$\langle e^{iM}\rangle = 1 - \mu_2/2 + r, \qquad |r| \le t\mu_2/6.$$

Since $\mu_2 \le t^2 \le 1/4$, the real main term is positive, and

$$\Big|\, |\langle e^{iM}\rangle| - (1 - \mu_2/2) \Big| \le t\mu_2/6.$$

Substitution into $\varepsilon(U)^2 = 2 - 2|\langle e^{iM}\rangle|$ proves (B.3) and its stated consequences. □

The constant $\sqrt{6/5} = 1.0954\ldots$ is therefore sharp to within about ten percent: the true asymptotic constant is $\sqrt{q}$, and the excess is the cubic Taylor cutoff, not structure.

Remark B.9 (The stability constant is a Fisher metric). For $U(s) = e^{isM}$, the pure-state quantum Fisher information is $4\,\mathrm{Var}_\psi(M)$ [BC94]. Hence (B.1) gives

$$F_Q = \frac{4}{q}\sum_{j=1}^{n} \|h_j\|_F^2.$$

Thus 2-uniformity makes the Fisher form a flat multiple of the Euclidean form on traceless one-site generators. This determines the local constant; it does not give a state-independent radius in the ambient product-unitary group.

Comparison with the main theorem. Theorem B.8 is a local estimate in a chosen logarithm chart and applies to nonstabilizer states. By contrast, Theorem 6.7 uses stabilizer cleaning to give an explicit state-space entry radius, select an exact Clifford branch, and control residual generators with spectral spread up to π. Thus the appendix explains the universal infinitesimal mechanism; it is not an alternative proof of the headline quantitative theorem.

C Weighted tests and the universal AME bound

This appendix refines Proposition 4.6. The first result quantifies the effect of selecting or reweighting tests. The second removes the stabilizer assumption from the uniform gap and weighted lower bound. Neither refinement is needed for the rigidity proofs or the observable criterion in Corollary 6.11.

For a perfectly complete verification operator Ω, write

$$\nu(\Omega) = \min_{v \perp \psi,\ \|v\|=1} \langle v|(I - \Omega)|v\rangle.$$

A permitted binary effect is $E_S \otimes I_{S^c}$, with $0 \le E_S \le I$, $|S| \le m+1$, and $(E_S \otimes I)\psi = \psi$. Randomization chooses among these fixed supports before testing the copy. Positivity implies that $I - E_S$ annihilates the marginal support. Consequently $E_S \otimes I \ge \Pi_S$ when $|S| = m+1$; for $|S| \le m$, full marginal rank forces $E_S = I$. This is why half-plus-one is the first informative support size.

Proposition C.1 (Weights and number of tests)**.** For a stabilizer AME$(2m, q)$, $m \ge 2$, assign probabilities w_S to the complementary library $\mathcal{S}$, including zero weights, and order them as $w_{(1)} \le \cdots \le w_{(2m)}$. Then

$$\nu\left(\sum_{S\in\mathcal{S}} w_S \Pi_S\right) = \sum_{a=1}^{m+1} w_{(a)}. \tag{C.1}$$

Among all permitted protocols with at most s binary tests,

$$\nu_{\max}(s) = \begin{cases} 0, & 1 \le s < m, \\ 1-(m-1)/s, & m \le s \le 2m, \\ (m+1)/(2m), & s \ge 2m. \end{cases} \tag{C.2}$$

Any s distinct library tests sampled uniformly attain the middle branch; the full library attains the last.

Proof. The character argument in Proposition 4.6 shows that every nontrivial character fails at least $m+1$ tests, and that any prescribed $m-1$ tests are exactly the pass set of some nontrivial character. Let it pass the $m-1$ heaviest weights to obtain (C.1).

For an arbitrary collection of permitted supports, each contributes at most $2e$ independent character constraints. Fewer than m tests therefore have a common nontrivial passing character. If there are $t \ge m$ positive weights, a nontrivial character passes the support projectors of the $m-1$ heaviest tests, and hence their effects. Its rejection probability is at most $1-(m-1)/t$. Combine this with the one-site-error cap $(m+1)/(2m)$ proved in the body. Uniform sampling from the library attains both bounds in their stated ranges. □

In particular, uniform weights uniquely maximize the gap within the full library. If $f \le m$ tests become unavailable, the remaining tests still give gap $(m+1-f)/(2m-f)$ when sampled uniformly. Here the parties remain available; only the list of tests is reduced. The general minimum-test gap bound and common-eigenbasis optimization precede this calculation [DHZ20, Proposition 1 and Section V.B].

Proposition C.2 (Arbitrary AME targets)**.** Let ψ be any AME$(2m, q)$, with integers $q \ge 2, m \ge 2$. For the same complementary library and arbitrary probability weights,

$$I - \sum_{S\in\mathcal{S}} w_S \Pi_S \ge \left(\sum_{a=1}^{m+1} w_{(a)}\right)(I - |\psi\rangle\langle\psi|). \tag{C.3}$$

Any m library marginals determine ψ among all density operators. The uniform full library has gap exactly $(m+1)/(2m)$, optimal among permitted tests. Exact weighted attainment and the finite-test upper bound above retain their stabilizer hypothesis.

Proof. We replace stabilizer characters by errors on a half-set. Choose on each site a unitary operator basis orthonormal for normalized Hilbert–Schmidt inner product, with identity distinguished. Maximal mixing on B makes the q^{2m} vectors $W_B\psi$ an orthonormal basis. The projector $\Pi_{C\cup\{i\}}$ has rank q^{2m-2}; it fixes the q^{2m-2} basis vectors whose error at i is identity, since those errors act on its complement. These vectors therefore span its range. Each star is consequently a commuting family, testing one error coordinate at a time, with product $|\psi\rangle\langle\psi|$.

Equivalently, a half-unitary taking ψ to m Bell pairs conjugates these projectors to the pair projectors. This is the virtual-factor picture of parent constraints [JTV17, Sections IV–V] [KJTV17, Sections 2 and 5].

Select r tests from this star and t from the other, where $s = r + t \geq m$, and put $d = s - m$. On $\psi^{\perp}$, let A, D be the respective sums of rejection projectors and let $E_k = \mathbf{1}_{A \leq k}$, $F_l = \mathbf{1}_{D \leq l}$. The range of E_k is spanned by nonidentity errors on B with at most k errors on the r tested coordinates, hence total weight at most $m - r + k$. The analogous bound for F_l is $m - t + l$. If $k + l \leq d$, the combined support of two such opposite-half errors has size at most m. Maximal mixing makes their inner product zero: the errors are on disjoint halves and each vector has a nonidentity factor of trace zero. Thus $E_k F_l = 0$.

In particular $E_k + F_{d-k} \leq I$ for $0 \leq k \leq d$. The spectra of A, D are nonnegative integers, so spectral calculus gives

$$\sum_{k=0}^{d} E_k = (d+1)I - \min\{A, (d+1)I\}.$$

Summing the projection inequalities yields $A + D \geq (d+1)I$. Restoring the target ray, every selected family of $s \geq m$ tests therefore satisfies

$$\sum_{S \text{ selected}} (I - \Pi_S) \geq (s - m + 1)(I - |\psi\rangle\langle\psi|).$$

Order the weights, write $w_0 = 0$, and expand the weighted sum as

$$\sum_{i=1}^{2m} w_i (I - \Pi_i) = \sum_{k=1}^{2m} (w_k - w_{k-1}) \sum_{i=k}^{2m} (I - \Pi_i).$$

Apply the selected-family bound to tails of size at least m, and positivity to the rest. The coefficient telescopes to $\sum_{i=1}^{m+1} w_i$, proving (C.3). A positive gap for any m selected tests gives marginal determination. Uniform full sampling attains the one-site-error upper bound from the body, whose proof uses only one-party maximal mixing. □

D Verification and trust boundary

The principal results are conceptual proofs. This paper contains no paper-facing finite census, generated certificate, or numerical experiment. Its trust boundary has three layers: complete manuscript proofs, selected kernel-checked algebraic cores, and standard results imported by citation.

Table 2: Trust boundary for the theorem families retained in this paper. "Core formalization" means that the named algebraic step is kernel checked; it does not promote the complete manuscript theorem to a formal theorem.

Result family	Manuscript proof	Formal status
AME support counting and complete Weyl-basis rigidity	Sections 2–3	coordinate squeeze, unique local labels, and the abstract axis criterion are kernel checked; the concrete additive-state marginal composition is not
Minimum-support transition maps and phase correction	Section 4	minimum-support generation, the abstract holonomy centralizer, and the stabilizer-character algebra are kernel checked; the state-ray correction is a manuscript step; the systematic half-set form, block-minor criterion, optimal direct sum, and finite recognition bounds are manuscript only
Marginal determination, complementary tests, and stochastic conversion	Section 4 and Appendix C	manuscript only; weighted and universal verification bounds have no formal coverage
Local endomorphism algebra, five prime-field types, and low-party theorem	Section 5 and Appendix A	manuscript only; the existing abstract holonomy-centralizer core does not formalize the intrinsic algebra identification, five-type composition, low-party commutativity, or module consequences
Encoder Clifford no-go	Section 4	the Choi orientation and Clifford closure are kernel checked; the complete arbitrary-additive encoder theorem is assembled in the manuscript
Cleaning, Weyl–Fourier rounding, branch selection, and residual stability	Section 6	manuscript only; neither the leakage constant 8, the Fourier rounding lemma, the balanced-cut estimate, nor the global radius is formalized
Relative exact-base intertwiner reduction	Section 6	exact line transport, defect invariance, product-intertwiner composition, and lossless transfer of a supplied threshold and coefficient are kernel checked; the cleaning inputs remain manuscript proofs
Robust transition compatibility and character obstruction	Section 6	manuscript only
Two-uniform discreteness and local stability	Appendix B	only the generator splitting, single-exponential identity, polarized second-moment identity, and absence of a nonscalar continuous symmetry are kernel checked; the local quadratic constants are manuscript only

Table 2 is a disclosure of partial formalization, not an independently reproducible formal artifact for the principal theorems. The selected cores were developed in the Lean namespace `RelativeConicArcs.AMELU`. The paper does not supply a self-contained replay contract matching its statements to a pinned public set of formal declarations, and claims no end-to-end formalization of any theorem family. All principal results are proved in the manuscript independently of these partial formal developments.

The cleaning theorem uses three cited exact inputs: stabilizer cleaning, localization of nested commutators, and the exact three-cleanable-region Clifford obstruction. The manuscript proves the leakage estimate, the finite Weyl–Fourier concentration step, the stabilizer overlap gap, and their AME composition. Its closed-form deductions do not depend on a computational search or floating-point evaluation.

The quantitative appendix serves as a mechanism comparison. None of its results is used to prove Theorem 1.2.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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